Braided module categories from small braidings

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Abstract

We introduce small braidings on inclusions of monoidal categories $\mathcal{C} \hookrightarrow \mathcal{D}$, which allow one to braid objects in $\mathcal{C}$ with objects in $\mathcal{D}$. A small braiding makes $\mathcal{C}$ a braided monoidal category, and $\mathcal{D}$, as well as certain subcategories of $\mathcal{D}$, a braided module category. We prove versions of previously known reconstruction results for braided module categories which do not rely on finiteness assumptions. We construct a number of new examples coming from quantum groups, and recover the quantum group equivariant quantisations of conjugacy classes of Lie groups studied by Donin and Mudrov.

Contents

1 Introduction **2**
1.1 Outlook 4
1.2 Organisation 5

2 Categorical preliminaries **6**
2.1 Braided module categories 6
2.2 Half-braided algebras and quantum moment maps 9
2.3 Small braidings 24

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3 Braided module categories for quantum groups **30**
3.1 Small braidings from unrolled quantum groups 30
3.2 Examples for generic quantum parameter 33
3.3 Quantum moment maps from internal-homs 35

A Closed monoidal categories **41**

B Derived I-adic completion **46**

1 Introduction

Braided monoidal categories can be thought of as monoidal categories which provide representations of the braid groups of Coxeter type A [JS93]. Adopting this intuition, *braided module categories*, first introduced in [En08], extend these representations to representations of braid groups of type B [Br13].

Braided module categories arise naturally in the Morita theory of braided monoidal categories. They are objects in the endomorphism category of the identity morphism in the 4-category of braided tensor categories [BJS18], see [Jo23, Sec. 3.3] for a proof of this, or objects in the centre of the category of module categories over a braided monoidal category [DN21, Thm. 4.11]. Braided module categories have provided a categorical interpretation of quantum K-matrices [Ko17, LWY23], and have found applications in quantum topology [BZBJ18b], and category theory [JFR23].

In this paper we present a new construction for braided module categories. Namely, let $\mathcal{C}$ and $\mathcal{D}$ be monoidal categories and let there be an inclusion of monoidal categories $\mathcal{C} \hookrightarrow \mathcal{D}$. We define a *small braiding* (Definition 2.18) which allows one to braid objects in $\mathcal{C}$ with objects in $\mathcal{D}$. A small braiding equips $\mathcal{C}$ with a braiding and any subcategory of $\mathcal{D}$ which is closed under tensoring with objects in $\mathcal{C}$ with the structure of a braided module category. An advantage of this construction is that it does not rely on finiteness or linearity assumptions for the categories in question. As far as the authors are aware, current systematic approaches for constructing braided module categories result in a category which is the category of finite-dimensional modules over an algebra [Ko17, LWY23], or locally finite-dimensional modules [BZBJ18b] (see also [BZBJ18a, Sec. 1.3.2]).

Let $\mathcal{C}$ and $\mathcal{D}$ be abelian categories and let the inclusion $\mathcal{C} \hookrightarrow \mathcal{D}$ be right exact. Given an algebra $A \in \mathcal{D}$, the category $\mathcal{D}_A$ of A-modules in $\mathcal{D}$ is a $\mathcal{D}$-module category. Furthermore, since $\mathcal{C}$ is a monoidal subcategory, $\mathcal{D}_A$ is also a $\mathcal{C}$-module category. If the algebra A is in the image of the inclusion $\mathcal{C} \hookrightarrow \mathcal{D}$, module braidings for $\mathcal{D}_A$ are known to be in bijection with half-braidings [JFR23, Sec. 2.5], and with quantum moment maps when $\mathcal{C}$ is ribbon [BZBJ18b, Cor. 4.11] (see Proposition 2.9

and Definition 2.13 for the definitions of these terms). If A is not in the image of the inclusion, we can still show one direction of the bijection: A half-braiding for A results in a module braiding for $\mathcal{D}_A$. It remains true that if $\mathcal{C}$ is ribbon, half-braidings for A are in bijection with quantum moment maps.

The definition of a small braiding is motivated by observing that the braiding for quantum groups $U_q\mathfrak{g}$ with generic quantum parameter q is well-defined on a tensor product of type 1 modules if at least one of the modules is finite-dimensional. We extend this result to braid generic weight modules with finite-dimensional type 1 modules by showing that there is a monoidal inclusion of the category of finite-dimensional type 1 modules for $U_q\mathfrak{g}$ into the category of weight modules for the unrolled quantum group $U_q^H\mathfrak{g}$. From this we obtain the following small braiding:

Theorem 1.1 (Theorem 3.4)**.** Let $\mathcal{C}$ be the category of finite-dimensional $U_q\mathfrak{g}$-modules of type 1 and $\mathcal{D}^H$ the category $U_q^H\mathfrak{g}$-weight modules. For $q \in \mathbb{C}^\times$ not a root of unity, there is a small braiding on the monoidal inclusion $\mathcal{C} \hookrightarrow \mathcal{D}^H$ given by

$$\overrightarrow{\sigma}_{U,V}(u \otimes v) = s_{U,V} q^{\sum_{i,j=1}^n d_i (A^{-1})_{ij} H_i \otimes H_j} R \cdot (u \otimes v) \tag{1.1a}$$

$$\overleftarrow{\sigma}_{V,U}(v \otimes u) = s_{V,U} q^{\sum_{i,j=1}^n d_i (A^{-1})_{ij} H_i \otimes H_j} R \cdot (v \otimes u)\ , \tag{1.1b}$$

where R is the R-matrix for $U_q\mathfrak{g}$ and s is the swap operation on vector spaces.

Using the above theorem and basic facts about the representation theory of $U_q^H\mathfrak{g}$ at generic quantum parameter q we describe several concrete examples coming from this small braiding:

- (Example 3.5) $\mathcal{O}_q^H$, the quantum category $\mathcal{O}$ of [AM14, Def. 3.1] for the unrolled quantum group.
- (Example 3.6) Given $\Lambda \in \mathfrak{h}^*/\sim$, where $\lambda \sim \mu$ if $\lambda - \mu$ is integral, one can take $\mathcal{D}_\Lambda^H$, the full subcategory of $\mathcal{D}^H$ of modules with weight support in Λ.
- (Example 3.8) For $\Lambda \in \mathfrak{h}^*/\sim$ containing only generic weights, $\mathcal{M}_\Lambda^H$ the full subcategory of $\mathcal{D}_\Lambda^H$ with objects isomorphic to a direct sum of Verma modules with highest weights in Λ.

In the Appendix we review how internal-end objects $\underline{\mathcal{D}}(V,V) =: \underline{\mathcal{D}}(V)$ in a monoidal category $\mathcal{D}$, if they exist, come with the canonical structure of an algebra, and how the internal-hom functor $\underline{\mathcal{D}}(V,-) : \mathcal{D} \to \mathcal{D}_{\underline{\mathcal{D}}(V)}$ comes with a canonical natural transformation $\Phi_{U,W} : \underline{\mathcal{D}}(V, U \otimes W) \to U \otimes \underline{\mathcal{D}}(V,W)$. In Lemma A.5 we show that $\Phi_{U,W}$ is an isomorphism if and only if U has a left dual. Hence $\underline{\mathcal{D}}(V,-)$ is a module functor if and only if we are considering $\mathcal{D}$ and $\mathcal{D}_{\underline{\mathcal{D}}(V,V)}$ as module categories over a left-rigid monoidal subcategory of $\mathcal{D}$. In the $U_q\mathfrak{g}$ setting

described above, $\mathcal{C}$ is precisely the full subcategory of left-rigid objects in $\mathcal{D}$. It is then natural to ask if internal-homs exist in $\mathcal{D}$, and whether we can equip them with quantum moment maps such that $\underline{\mathcal{D}}(V,-)$ becomes a braided module functor.

We are able to construct quantum moment maps from the $\hbar$-adic Hopf algebra $U_\hbar\mathfrak{g}$, where $q=e^\hbar$ and $\hbar$ is now a formal parameter instead of a complex number. We use the fact that the category $\mathcal{D}$ of topologically free modules is closed monoidal to construct quantum moment maps for internal endomorphism algebras $\underline{\mathcal{D}}(V)$. Because the representation theory of topologically free $U_\hbar\mathfrak{g}$ weight modules mimics that of $U_q\mathfrak{g}$-weight modules, we are also able to work with the above examples in the $\hbar$-adic setting. We then construct half-braidings using a small braiding between the category of finite-rank topologically free $U_\hbar\mathfrak{g}$-modules and $\mathcal{D}$. Since $\mathcal{C}$ is ribbon, we arrive at the following theorem.

Theorem 1.2. (Theorem 3.12) Let $\mathcal{D}$ be the category of derived $I_\hbar$-adic complete $U_\hbar\mathfrak{g}$-modules and $\mathcal{C}$ be the category of finite-rank topologically free $U_\hbar\mathfrak{g}$-modules. For all $V\in\mathcal{D}$ there is a canonical quantum moment map for the internal-endomorphism algebra $\underline{\mathcal{D}}(V)$ of V. Taking V to be a generalised Verma module, the quantum moment map for $\underline{\mathcal{D}}(V)$ recovers the quantised conjugacy classes of Lie groups defined in [Mu07].

In Proposition 3.13 we observe that the braided module category $\mathcal{M}_\Lambda$ satisfies sufficient conditions (Proposition A.7) such that the functor $\underline{\mathcal{D}}(V(\lambda),-):\mathcal{M}_\Lambda\to\mathcal{D}_{\underline{\mathcal{D}}(V(\Lambda))}$ is a fully faithful (but not essentially surjective) braided $\mathcal{C}$-module functor.

1.1 Outlook

Our main motivation comes from an application in defect skein theory. Namely, organising defect data of skein modules according to the Morita theory of braided tensor categories [BJS18], one is is lead to label line defects with braided module categories and junctions between line defects with braided module functors. In future work, the authors intend to study examples of defect skein modules with defect data coming from the examples discussed here. See [BZBJ18b, BJ25, G26, KR26, BGJM26, CJR26] for previous work of a similar flavour on defect skein modules.

By the final observation, Remark 3.14, we show why the inclusion of braided module categories $\mathcal{M}_\Lambda\hookrightarrow\mathcal{D}_{\underline{\mathcal{D}}(V(\lambda)}$ is not an equivalence. Tools for monadic reconstruction of module categories were developed in [SZ24a] precisely in order to deal with non-finite and non-rigid settings. Any $V(\lambda)\in\mathcal{M}_\Lambda$ is a $\mathcal{C}$-projective $\mathcal{C}$-generator in the sense of [SZ24a, Def. 4.2]. Hence if one can relax the conditions on $\mathcal{C}$ and $\mathcal{M}$ being abelian in [SZ24a, Thm. 5.1], it may be possible for $\mathcal{M}_\Lambda$ to be reconstructed monadically. To reconstruct the braiding, one requires a monadic version of half-braidings/quantum moment maps.

The quantum group setting is amenable to constructing many more examples of braided module categories from small braidings. One could study the tensor product of finite-dimensional modules with generlised Verma modules in [Mu07, Prop. 3.7] to construct braided module categories similar to $\mathcal{M}_\Lambda$. These examples should also have counterparts when the quantum parameter is specialised to a complex number. Examples beyond highest weight modules could be constructed by studying the tensor product of the torsion-free modules of [FHW15] with finite-dimensional modules. Furthermore, all of these examples will admit quantum moment maps in the $\hbar$-adic setting, which provide more quantum group equivariant coadjoint orbits of Lie groups à la [DM02, Mu07].

The example $\mathcal{M}_\Lambda$ for $\mathfrak{g} = \mathfrak{sl}_2$ has appeared under various guises in the literature, without the braided module category structure being recognised: The category $\mathcal{M}_\Lambda$ describes boundary conditions for the XXZ spin chain [CGS22], one can find in its morphism spaces the blob algebra [MS94], and morphism spaces of the blob category of [ILZ19, LNV21]. It would be interesting to explore these connections more deeply.

1.2 Organisation

In Section 2.1 we review the basics of braided module categories and braided module functors. Then in Section 2.2 we review the construction of braided module categories via half-braided algebras and quantum moment maps, and braided module functors via half-braided bimodules. We then introduce small braidings in Section 2.3, and establish generalised versions of half-braided algebras, quanutum moment maps, and half-braided bimodules in this setting. In Section 3 we study braided module categories coming from quantum groups. We begin by discussing the small braiding between finite-dimensional type 1 modules for $U_q\mathfrak{g}$ and weight modules for $U_q^H\mathfrak{g}$ in Section 3.1. We then introduce concrete examples in this setting in Section 3.2. We conclude in Section 3.3 by switching settings to the $\hbar$-adic quantum group, where we construct half-braided algebras and quantum moment maps using the internal-homs. In Appendix A we prove some results about algebras in closed categories which we use in Section 3.3.

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AI usage disclosure

During the preparation of this manuscript we used Claude to search for relevant literature, as well as to proof-read a draft during the final editing. None of the text or diagrams in this manuscript are generated by AI. We hold full responsibility for the validity of the results in this manuscript.

2 Categorical preliminaries

In this section we review the notion of a braided module category over a braided monoidal category and its corresponding graphical calculus. We then review the results of [BZBJ18b] and [JFR23] which construct braided module categories from so-called quantum moment maps and half-braided algebras respectively. Finally, we define small braidings, and develop the analogues of quanutum moment maps and half-braided algebras in this new setting.

Given a category $\mathcal{C} \ni U, V$, we denote the space of morphisms $U \to V$ as $\mathcal{C}(U, V)$. for Our convention for a braided monoidal category $(\mathcal{C}, \otimes, \mathbb{1}, \alpha, l, r, \sigma)$ follow those of [Ka95, Ch.s XI & XII]. Namely, the direction of the associator is

$$\alpha : (- \otimes =) \otimes = \Rightarrow - \otimes (= \otimes =) \tag{2.1}$$

Our convention and notation for a rigid monoidal category is the same as that of [EGNO15, Sec. 2.10]. We define a ribbon category as in [EGNO15, Def. 8.10.1]. We abbreviate the idenitity morphism on objects id_V by simply writing the object V when there is no ambiguity in doing so.

We frequently employ a digrammatic calculus for braided and rigid monoidal categories. Diagrams are to be read from the bottom up, with the structural morphisms being depicted

$\sigma_{U,V} =$  , $\mathrm{ev}_V =$  , $\mathrm{coev}_V =$  . (2.2)

2.1 Braided module categories

Let $\mathcal{C}$ be a braided monoidal category.

Definition 2.1. ([EGNO15, Def. 7.1.2]) A *(left) $\mathcal{C}$-module category* is a category $\mathcal{M}$ along with an *action bifunctor* $- \triangleright = : \mathcal{C} \times \mathcal{M} \to \mathcal{M}$ and a *modulator*, which is a natural isomorphism

$$\alpha^{\mathcal{M}} : (- \otimes =) \triangleright \equiv \Rightarrow - \triangleright (= \triangleright \equiv) , \tag{2.3}$$

as well as a unitor isomorphism $r^{\mathcal{M}} : \mathbb{1} \triangleright - \Rightarrow \mathrm{id}_{\mathcal{M}}$. The modulator is required to satisfy the mixed pentagon identity, and the unit isomorphism the mixed triangle identity, both of which can be found in [EGNO15, Sec. 7.1].

Example 2.2. ([EGNO15, Sec. 7.8]) Let (A, m, u) be an algebra in $\mathcal{C}$. The category $\mathcal{C}_A$ of right A-modules in $\mathcal{C}$ is a left $\mathcal{C}$-module category.

Definition 2.3. A *$\mathcal{C}$-module functor* $F : (\mathcal{M}, \triangleright) \to (\mathcal{N}, \triangleright)$ is a functor $F : \mathcal{M} \to \mathcal{N}$ together with a natural isomorphism $F_{-,=} : - \triangleright F(=) \Rightarrow F(- \triangleright =)$ such that the diagram

$$\begin{array}{ccc} & (U \otimes V) \triangleright F(M) & \\ {\scriptstyle \alpha^{\mathcal{N}}}\swarrow & & \searrow{\scriptstyle F_{U\otimes V,M}} \\ U \triangleright (V \triangleright F(M)) & & F((U \otimes V) \triangleright M) \\ {\scriptstyle U\triangleright F_{V,M}}\downarrow & & \downarrow{\scriptstyle F(\alpha^{\mathcal{M}})} \\ U \triangleright F(V \triangleright M) & \xrightarrow{\quad F_{U,V\triangleright M} \quad} & F(U \triangleright (V \triangleright M)) \end{array} \tag{2.4}$$

commutes for all objects U, V in $\mathcal{C}$ and M in $\mathcal{M}$.

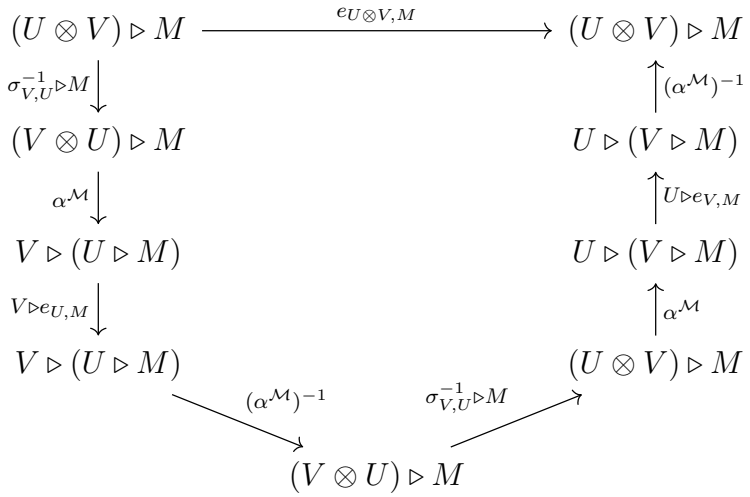
Example 2.4. ([EGNO15, Sec. 7.8]) Assume $\mathcal{C}$ is in addition abelian and that the tensor product is right exact. Let (A, m, u) and (B, m', u') be algebras in $\mathcal{C}$ and M be an (A, B)-bimodule in $\mathcal{C}$. The functor $- \otimes_A M : \mathcal{C}_A \to \mathcal{C}_B$ is a $\mathcal{C}$-module functor.

Definition 2.5. A *(left) braided $\mathcal{C}$-module category* is a $\mathcal{C}$-module category $(\mathcal{M}, \triangleright, \alpha^{\mathcal{M}}, r^{\mathcal{M}})$ along with a *module braiding*, which is a natural automorphism of the action functor

$$e_{-,=} : - \triangleright = \to - \triangleright = , \tag{2.5}$$

such that the diagrams

$$\begin{array}{ccccccc} (U \otimes V) \triangleright M & \xrightarrow{\alpha^{\mathcal{M}}} & U \triangleright (V \triangleright M) & \xrightarrow{e_{U,V\triangleright M}} & U \triangleright (V \triangleright M) & \xrightarrow{(\alpha^{\mathcal{M}})^{-1}} & (U \otimes V) \triangleright M \\ {\scriptstyle \sigma_{U,V}\triangleright M}\downarrow & & & & & & \uparrow{\scriptstyle \sigma_{V,U}\triangleright M} \\ (V \otimes U) \triangleright M & \xrightarrow[\alpha^{\mathcal{M}}]{} & V \triangleright (U \triangleright M) & \xrightarrow[V\triangleright e_{U,M}]{} & V \triangleright (U \triangleright M) & \xrightarrow[(\alpha^{\mathcal{M}})^{-1}]{} & (V \otimes U) \triangleright M \end{array} \tag{2.6a}$$



(2.6b)

commute for all $M \in \mathcal{M}$ and $X, Y \in \mathcal{C}$.

Note that naturality of the associativity ismorphisms and condition (2.6b) for $U = V = \mathbb{1}$ implies $e_{\mathbb{1},M} = e_{\mathbb{1},M} \circ e_{\mathbb{1},M}$. Since $e_{\mathbb{1},M}$ is an isomorphism, $e_{\mathbb{1},M} = \mathrm{id}_M$.

Remark 2.6. Braided module categories were first introduced in [En08]. Several different conventions have emerged regarding their definition, the complete list (to the authors' knowledge) being [En08, Sec. 4.3], [Br13, Sec. 5.1], [BZBJ18b, Def. 3.4], [Ko17, Def. 2.1], [DN21, Def. 4.1], [JFR23, Sec. 2.2] and [LWY23, Def. 3.5]. We choose the same convention as [LWY23, Def. 3.5], allowing for $\mathcal{C}$ to be a braided module category over itself with module braiding $e_{U,V} = \sigma_{V,U} \circ \sigma_{U,V}$. Every definition listed is equivalent when the appropriate assumptions are made on $\mathcal{C}$, see [LWY23, Rem. 3.15] for a detailed comparison.

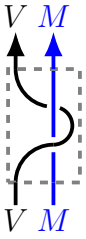
Remark 2.7. It is a standard result that any monoidal category is monoidally equivalent to a strict monoidal category (see for example [EGNO15, Thm. 2.8.5]). A similar result applies for module categories: A $\mathcal{C}$-module category is equivalent to a module category over the strictification of $\mathcal{C}$ such that the mixed associator and unitor are identities [Br13, Section 5.1]. We call such a module category a *strict module category*.

Definition 2.8. A $\mathcal{C}$-module functor $F : (\mathcal{M}, \triangleright) \to (\mathcal{N}, \triangleright')$ of braided module categories $(\mathcal{M}, e)$ and $(\mathcal{N}, e')$ is *braided* if the diagram

$$\begin{array}{ccc} V \triangleright' F(M) & \xrightarrow{e'_{V,F(M)}} & V \triangleright' F(M) \\ {\scriptstyle F_{V,M}}\downarrow & & \downarrow{\scriptstyle F_{V,M}} \\ F(V \triangleright M) & \xrightarrow{F(e_{V,M})} & F(V \triangleright M) \end{array} \tag{2.7}$$

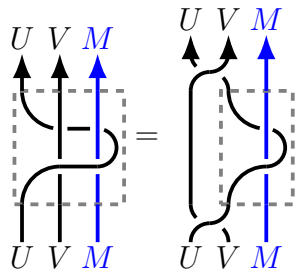
commutes for all $V \in \mathcal{C}$, $M \in \mathcal{M}$.

Let $\mathcal{M}$ be a braided $\mathcal{C}$-module category. We use the following graphical notation for the module braiding.

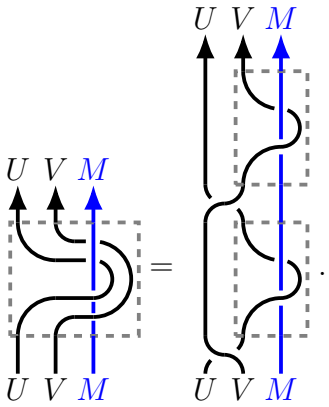
$e_{V,M} =$ 

(2.8)

The graphical notation for a braided module category implicitly uses the equivalence to a strict braided module category over a strict braided monoidal category. The conditions (2.6a) and (2.6b) are quite natural in the graphical notation,



(2.9a)



(2.9b)

2.2 Half-braided algebras and quantum moment maps

Let $\mathcal{C}$ be a braided monoidal category. In [JFR23] the theory of half-braided algebras, including the results of this section, was developed under the assumption that $\mathcal{C}$ is a fusion category. However, neither semisimplicity, rigidity, nor finiteness are required in the proofs of the statements we use. We include the proofs below as we refer to them in the sequel.

Proposition 2.9. [JFR23, Lemma 2.38] Let (A, m, u) be an algebra in $\mathcal{C}$. There is a bijective correspondence between module braidings for $\mathcal{C}_A$, and half-braidings

$\gamma^A : - \otimes A \Rightarrow A \otimes -$, depicted

$$\gamma_V^A = \quad , \tag{2.10}$$

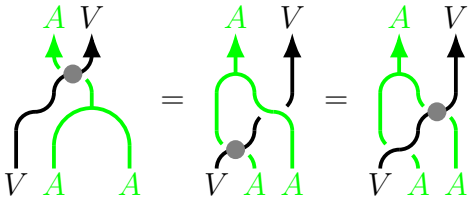


satisfying the hexagon [EGNO15, Eq. (7.41)], such that the diagram

$$\begin{array}{ccccc}
 & & (A \otimes V) \otimes A & \xrightarrow{(A \otimes \sigma_{A,V}^{-1}) \circ \alpha} & A \otimes (A \otimes V) \\
 & \nearrow^{\gamma_V^A \otimes A} & & & \searrow^{(m \otimes V) \circ \alpha^{-1}} \\
(V \otimes A) \otimes A & & \xrightarrow{(V \otimes m) \circ \alpha} & V \otimes A \xrightarrow{\gamma_V^A} & A \otimes V \\
 & \searrow_{\sigma_{V,A} \otimes A} & & & \nearrow_{(m \otimes V) \circ \alpha^{-1}} \\
 & & (A \otimes V) \otimes A & \xrightarrow[(A \otimes \gamma_V^A) \circ \alpha]{} & A \otimes (A \otimes V)
\end{array} \tag{2.11}$$

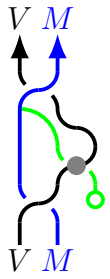
commutes for all objects V in $\mathcal{C}$. The condition (2.11) is depicted

$$= \quad = \tag{2.12}$$

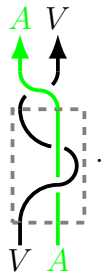


Proof. The proof is essentially identical to that of [JFR23, Lemma 2.38], although our convention differs from theirs. We reproduce it string-diagrammatically. We show via a direct computation that a module braiding can be obtained from a half-braiding via

$$e_{V,M} := \quad , \tag{2.13}$$

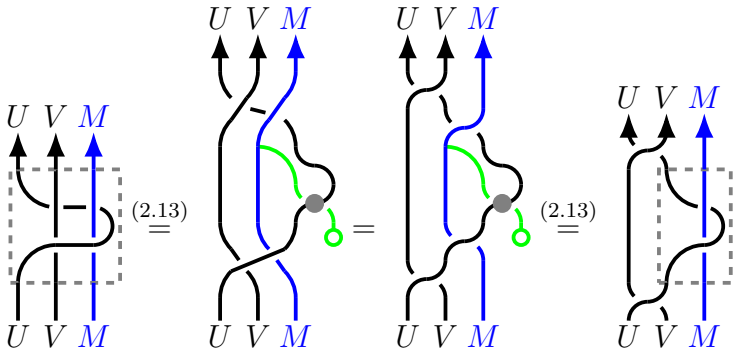


and a half-braiding from a module braiding via

$\gamma_V^A :=$ 

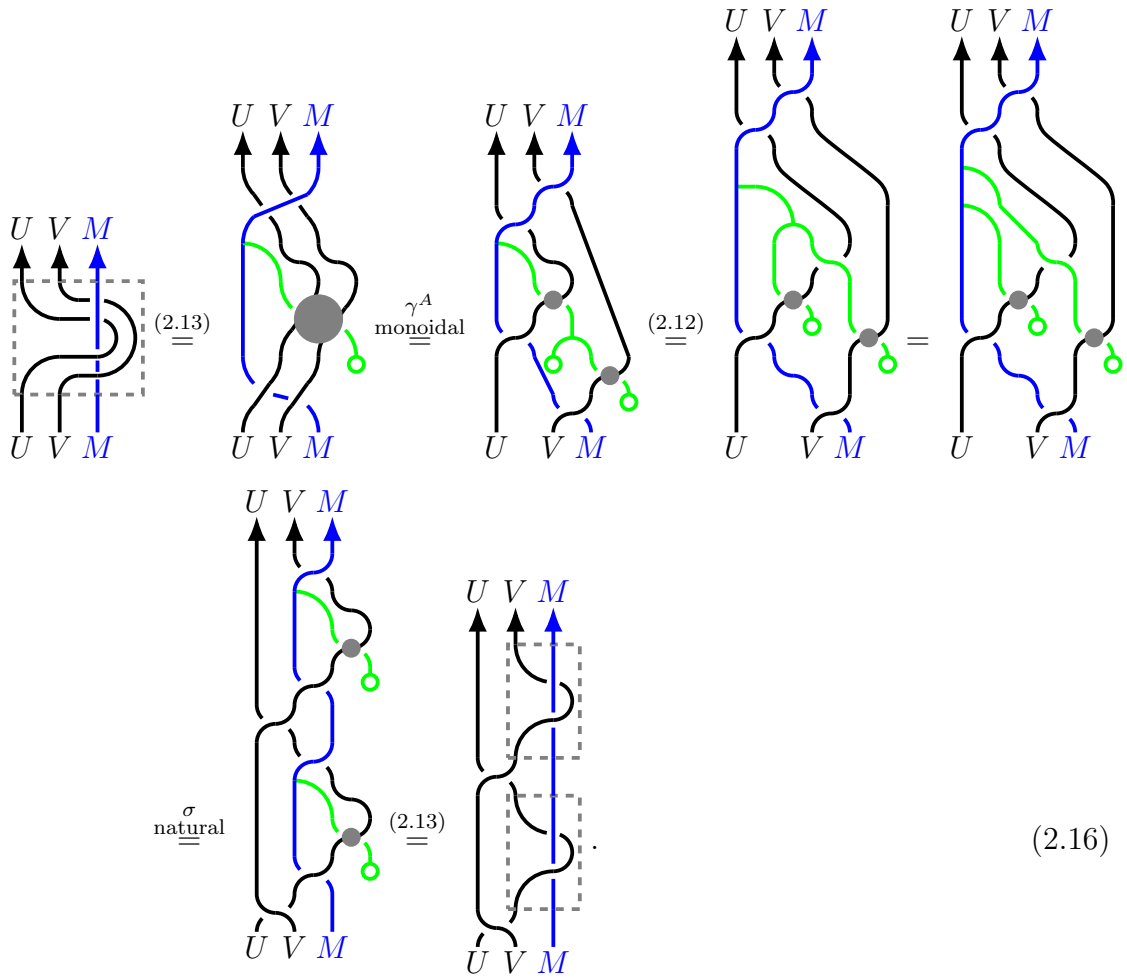
. (2.14)

Let U, V be objects in $\mathcal{C}$ and (M, ρ) an object in $\mathcal{C}_A$. Note that $V \triangleright M = (V \otimes M, V \otimes \rho)$ in the strict setting. First, we show that $e_{-,=}$ defined in (2.13) satisfies the condition (2.9a):



(2.15)

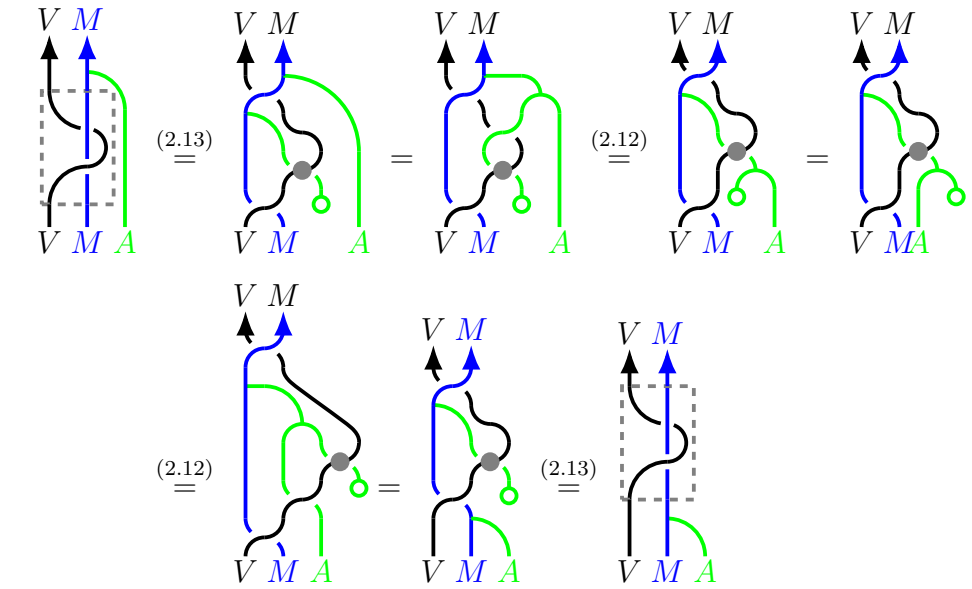
Now for the condition (2.9b):



(2.16)

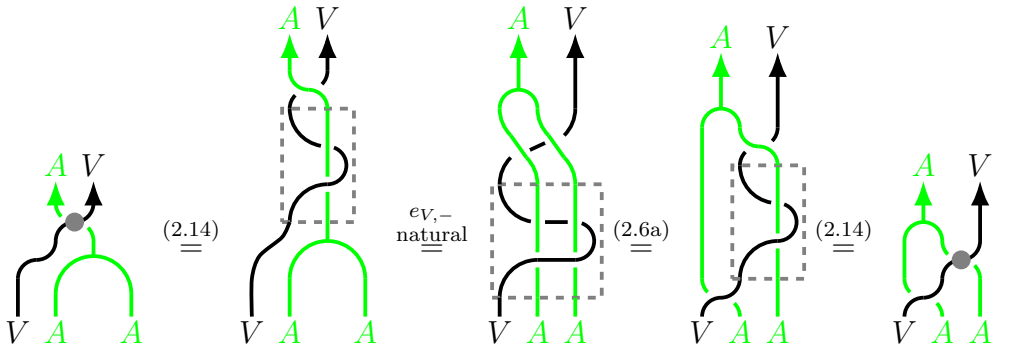
Finally, we show that each $e_{V,M}$ is a morphism in $\mathcal{C}_A$ (i.e. an intertwiner of A-

modules):

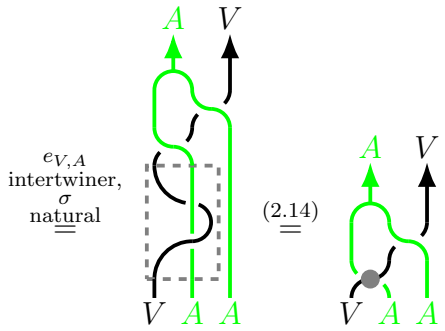


(2.17)

Now let us show that γ^A in (2.14) is a half-braiding.



(2.18)



(2.19)

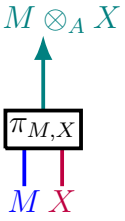
Showing that the assignments (2.13) and (2.14) are mutually inverse is a straight-

forward computation using the properties of half-braidings and module braidings. □

Definition 2.10. [JFR23, Def. 2.31] An algebra (A, m, u) in $\mathcal{C}$ equipped with a half-braiding γ^A satisfying (2.12) is called a *half-braided algebra*.

Remark 2.11. Setting $A = \mathbb{1}$ with the canonical (trivial) algebra we observe that a half-braiding for the regular $\mathcal{C}$-module category is equivalent to a monoidal natural automorphism of the identity functor, recovering [HS26, Prop. 2.2].

Let A and B be algebras in $\mathcal{C}$, M a right A-module in $\mathcal{C}$ and X an (A, B)-bimodule in $\mathcal{C}$. We depict the projector onto the relative tensor product $\pi_{M,X} : M \otimes X \to M \otimes_A X$ as

$\pi_{M,X} =$ 

(2.20)

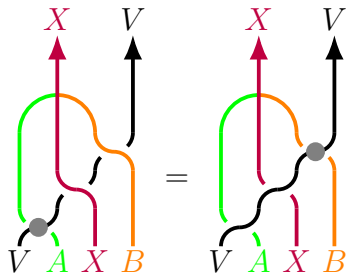
Proposition 2.12. [JFR23, Lem. 2.38] Let $((A, \gamma^A), m_A, u_A)$ and $((B, \gamma^B), m_B, u_B)$ be two half-braided algebras, and (X, ρ_A, ρ_B) be an (A, B)-bimodule in $\mathcal{C}$. The functor

$$- \otimes_A X : \mathcal{C}_A \to \mathcal{C}_B \tag{2.21}$$

is a braided module functor if and only if the diagram[1]

$$\begin{array}{ccccc}
((V \otimes A) \otimes X) \otimes B & \xrightarrow{(\gamma^A_V \otimes X) \otimes B} & ((A \otimes V) \otimes X) \otimes B & \xrightarrow{A \otimes \sigma^{-1}_{X \otimes B, V}} & A \otimes ((X \otimes B) \otimes V) \\
\downarrow{\scriptstyle \sigma_{V, A \otimes X} \otimes B} & & & & \downarrow{\scriptstyle (\rho_A \circ \rho_B) \otimes V} \\
(A \otimes (X \otimes V)) \otimes B & \xrightarrow[A \otimes (X \otimes \gamma^B_V)]{} & A \otimes (X \otimes (B \otimes V)) & \xrightarrow[(\rho_A \circ \rho_B) \otimes V]{} & X \otimes V
\end{array} \tag{2.22}$$

commutes for all $V \in \mathcal{C}$. The condition (2.22) is depicted graphically as



(2.23)

.

[1] We have suppressed the coherence morphisms for brevity.

Proof. Right exactness of the tensor product ensures the functor $-\otimes_A X : \mathcal{C}_A \to \mathcal{C}_B$ is a $\mathcal{C}$-module functor [EGNO15, Prop. 7.11.1]. To be a braided module functor it needs to satisfy the condition (2.7). We show this by showing that the diagram

$$\begin{array}{ccccc} V\otimes(M\otimes X) & \xrightarrow{V\otimes\pi_{M,X}} & V\otimes(M\otimes_A X) & \xrightarrow{e'_{V,(M\otimes_A X)}} & V\otimes(M\otimes_A X) \\ & & \sim\downarrow & & \downarrow\sim \\ & & (V\otimes M)\otimes_A X & \xrightarrow{e_{V,M}\otimes_A X} & (V\otimes M)\otimes_A X \end{array} \tag{2.24}$$

commutes for all $V \in \mathcal{C}$ and $M \in \mathcal{C}_A$. The $\mathcal{C}$-module functor structure, which is $\sim$ in (2.24), is the unique natural isomorphism $\phi_{-,=} : (-\otimes=)\otimes_A X \Rightarrow -\otimes(=\otimes_A X)$ such that

$$\begin{array}{ccc} V\otimes(M\otimes X) & \xrightarrow{V\otimes\pi_{M,X}} & V\otimes(M\otimes_A X) \\ \uparrow\alpha & & \phi_{V,M}\uparrow \\ (V\otimes M)\otimes X & \xrightarrow[\pi_{V\otimes M,X}]{} & (V\otimes M)\otimes_A X \end{array} \tag{2.25}$$

commutes. Hence (2.24) can be written,

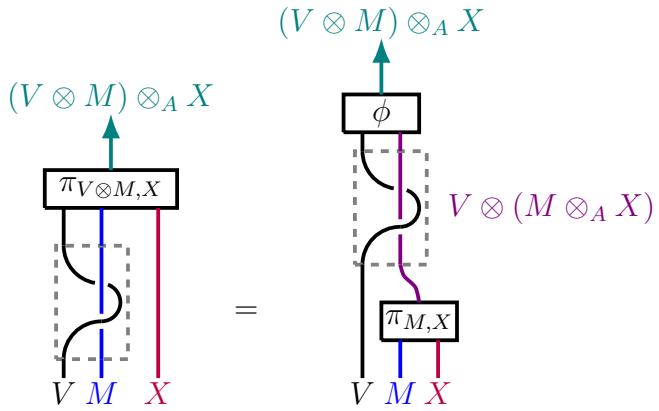
$$\begin{array}{ccccc} V\otimes(M\otimes X) & \xrightarrow{V\otimes\pi_{M,X}} & V\otimes(M\otimes_A X) & \xrightarrow{e'_{V,(M\otimes_A X)}} & V\otimes(M\otimes_A X) \\ \alpha^{-1}\downarrow & & \phi^{-1}_{V,M}\downarrow & & \downarrow\phi^{-1}_{V,M} \\ (V\otimes M)\otimes X & \xrightarrow[\pi_{V\otimes M,X}]{} & (V\otimes M)\otimes_A X & \xrightarrow{e_{V,M}\otimes_A X} & (V\otimes M)\otimes_A X \end{array} \tag{2.26}$$

where the left square commutes. Naturality of π implies (2.26) is equivalent to

$$\begin{array}{ccccc} V\otimes(M\otimes X) & \xrightarrow{V\otimes\pi_{M,X}} & V\otimes(M\otimes_A X) & \xrightarrow{e'_{V,(M\otimes_A X)}} & V\otimes(M\otimes_A X) \\ \alpha^{-1}\downarrow & & & \searrow^{\phi^{-1}_{V,M}} & \downarrow\phi^{-1}_{V,M} \\ (V\otimes M)\otimes X & \xrightarrow{e_{V,M}\otimes X} & (V\otimes M)\otimes X & \xrightarrow[\pi_{V\otimes M,X}]{} & (V\otimes M)\otimes_A X \end{array} \tag{2.27}$$

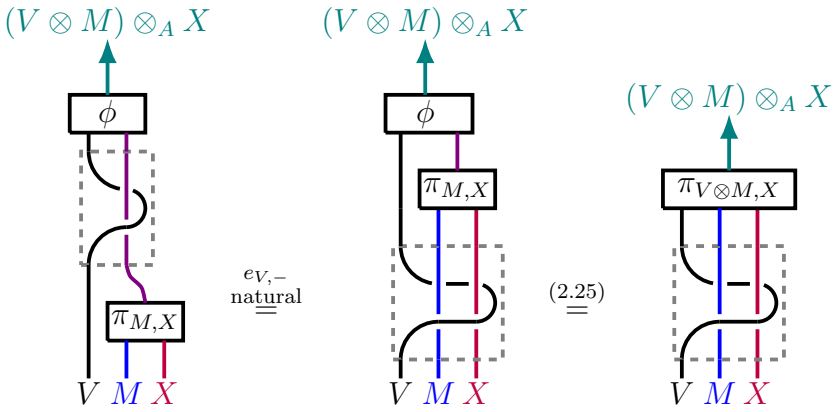
where the pentagon commutes. We now show that the outside rectangle of (2.27) commutes, which implies (2.24) commutes. The outside rectangle in (2.27) is

depicted



. (2.28)

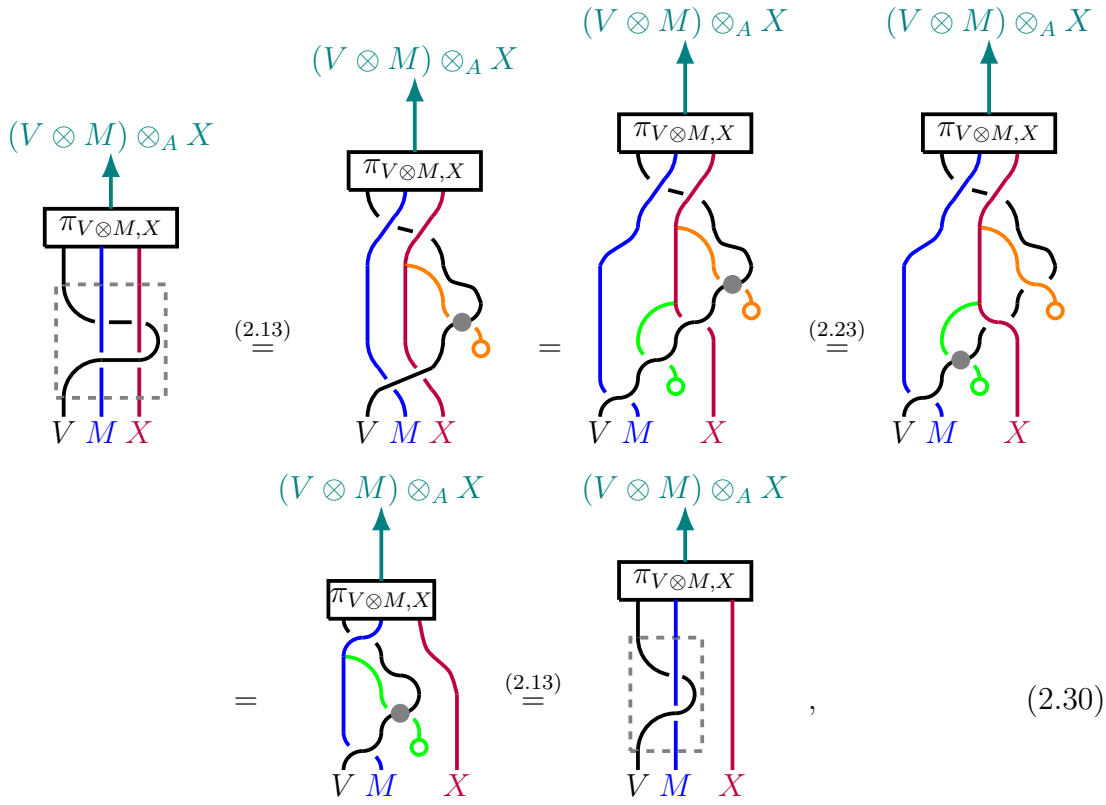
Using the fact that the module braiding is natural in each variable and $\pi_{M,X}$ is an intertwiner of right B-modules, we rewrite the right-hand side of (2.28) as



. (2.29)

We show that the left-hand side of (2.28) is equal to the right-hand side of (2.29)

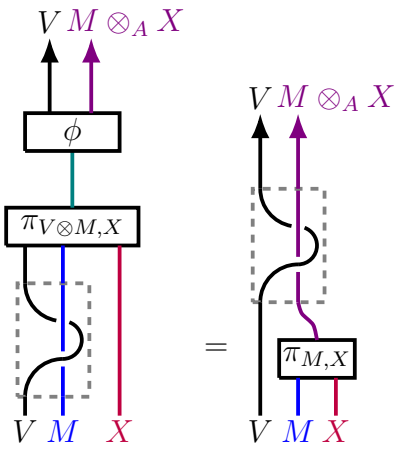
in the following string diagram computation.



(2.30)

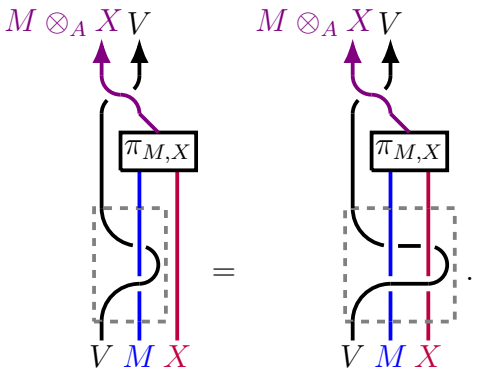
where in the second-last equality we used the presence of the projector onto $(V \otimes M) \otimes_A X$, and naturality of the braiding.

To prove the converse, note that $-\otimes_A N$ being a braided module functor implies



(2.31)

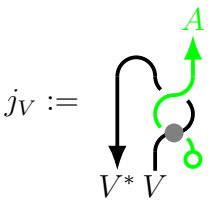
for all $M \in \mathcal{C}_A$. This in turn implies



(2.32)

From this point, a similar computation as in the previous direction results in (2.22). □

Assume $\mathcal{C}$ and $\mathcal{D}$ are abelian categories with a right exact tensor product, and that $\mathcal{C}$ is ribbon. From a half-braiding for an algebra $A \in \mathcal{C}$ we obtain a dinatural transformation $j : (-)^* \otimes = : \dddot{\to} A$ as follows:

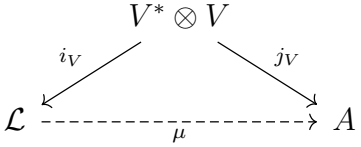


(2.33)

The universal recipient of dinatural transformations is the coend of the functor $(-)^* \otimes = : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{C}$, and is denoted

$$\mathcal{L} = \int^{V \in \mathcal{C}} V^* \otimes V \,. \tag{2.34}$$

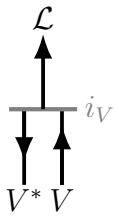
See [KL01, Ch. 5] for a detailed review of coends in braided monoidal categories. Assume $\mathcal{L}$ exists as an object in $\mathcal{C}$. By definition, it comes equipped with a dinatural transformation $i = (i_V : V^* \otimes V \to \mathcal{L})_{V \in \mathcal{C}}$ satisfying the following universal property: Given $A \in \mathcal{C}$ and dinatural transformation $j : (-)^* \otimes = : \dddot{\to} A$, there exists a unique morphism $\mu : \mathcal{L} \to A$ in $\mathcal{C}$ such that the diagram



(2.35)

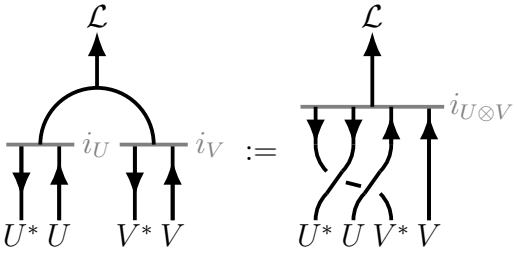
commutes for all V in $\mathcal{C}$. We depict the dinatural transformation i graphically as

$$i_V = \tag{2.36}$$

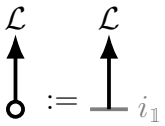


The morphisms $m : \mathcal{L} \otimes \mathcal{L} \to \mathcal{L}$ and $u : \mathbb{1} \to \mathcal{L}$ defined via

$$m \circ (i_U \otimes i_V) = \quad := \quad , \tag{2.37a}$$



$$u = \quad := \tag{2.37b}$$

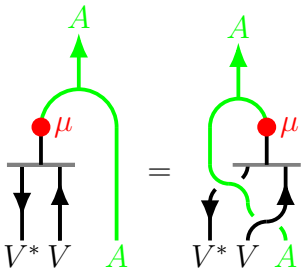


make $(\mathcal{L}, m, u)$ an algebra in $\mathcal{C}$.

Definition 2.13. Assume $\mathcal{C}$ is ribbon and contains the coend $\mathcal{L}$. A *quantum moment map* is an algebra homomorphism $\mu : \mathcal{L} \to A$ in $\mathcal{C}$ such that

$$\begin{array}{ccccc} (V^* \otimes V) \otimes A & \xrightarrow{(\mu \circ i_V) \otimes A} & A \otimes A & \xrightarrow{\quad m \quad} & A \\ {\scriptstyle (V^* \otimes \sigma_{V,A}) \circ \alpha}\downarrow & & & & \uparrow{\scriptstyle m} \\ V^* \otimes (A \otimes V) & \xrightarrow[(\sigma^{-1}_{A,V^*} \otimes V) \circ \alpha^{-1}]{} & A \otimes (V^* V) & \xrightarrow[A \otimes (\mu \circ i_V)) \circ \alpha]{} & A \otimes A \end{array} \tag{2.38}$$

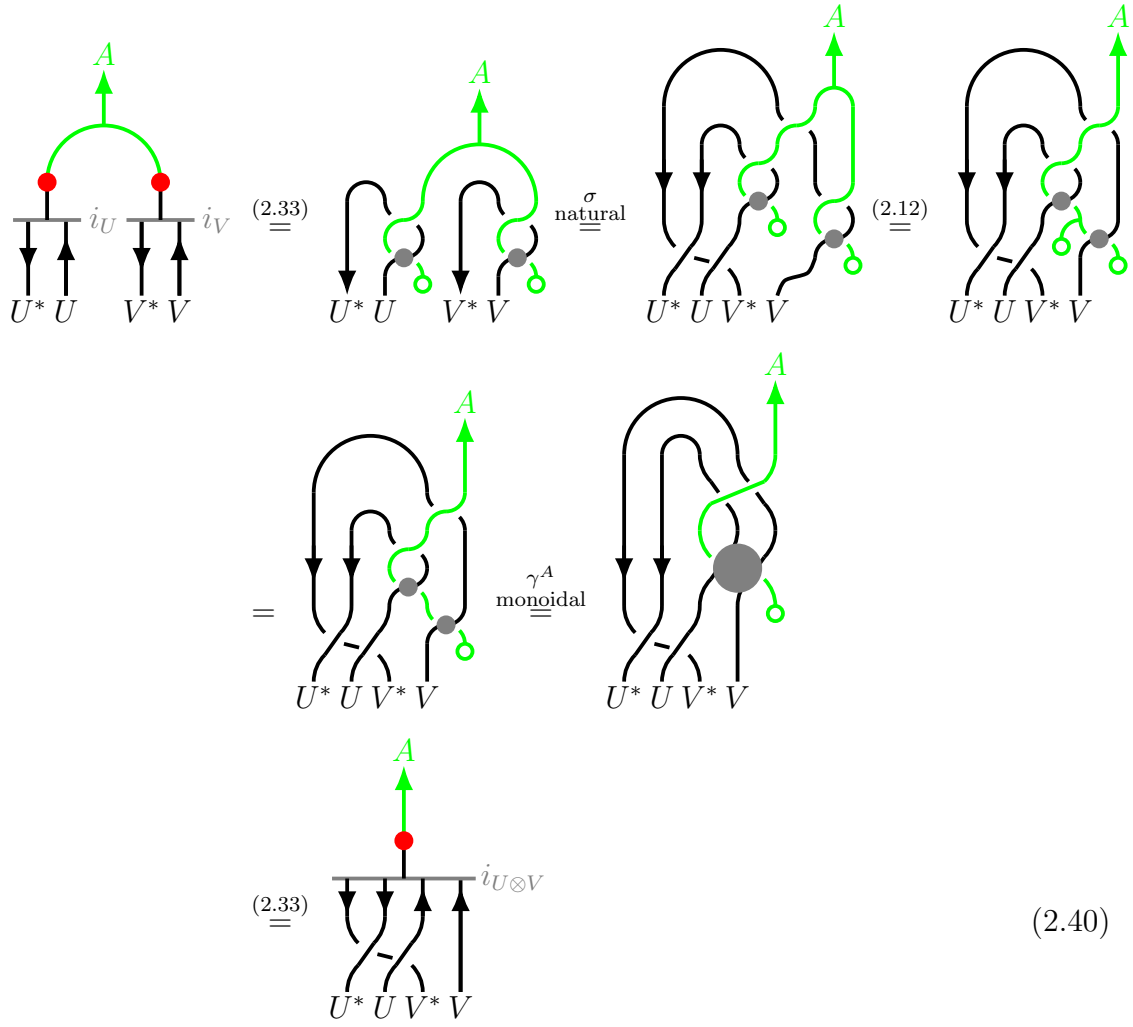
for all $V \in \mathcal{C}$. This condition is depicted



(2.39)

Proposition 2.14. [JFR23, Sec. 2.5] Assume $\mathcal{C}$ is ribbon and contains the coend $\mathcal{L}$. Given an algebra $A \in \mathcal{C}$, there is a bijective correspondence between half-braidings and quantum moment maps for A.

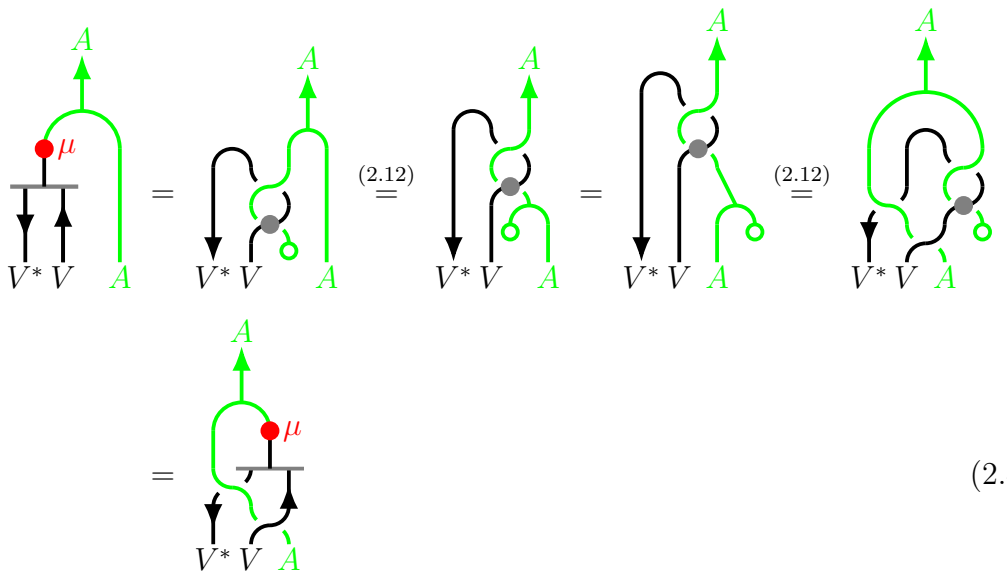
Proof. First we show that a half braiding can be used to construct a quantum moment map. From (2.33) we can construct a dinatural transformation from a half-braiding, and the universal property of the coend implies that there exists a unique morphism $\mu : \mathcal{L} \to A$. We now show that μ obtained this way is a quantum moment map. First, we show that it is an algebra homomorphism:



(2.40)

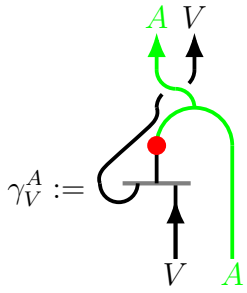
Preservation of the unit is clear from $\gamma^A_{\mathbb{1}} = \sigma_{A,\mathbb{1}} = \mathrm{id}_A$. Now we show that the

quantum moment map condition (2.39) is satisfied:



(2.41)

Now we consider the other direction. Given a quantum moment map $\mu : \mathcal{L} \to A$, we define a natural transformation $\gamma^A : - \otimes A \Rightarrow A \otimes -$ via

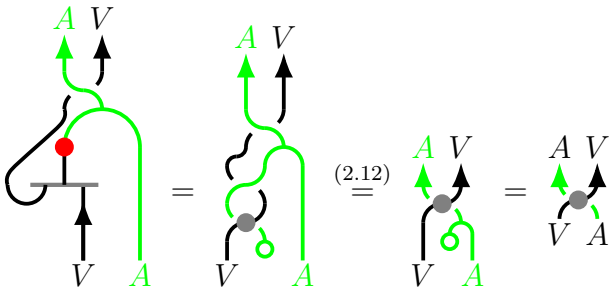


(2.42)

Dinaturality of $i : (-)^* \otimes = \ddot{\to} \mathcal{L}$ and naturality of the pivotal structure imply γ^A is natural. A straightforward string diagram computation shows that γ^A satisfies the conditions of a half braiding from Definition 2.9.

Finally, we show that these constructions are mutually inverse by inserting the quantum moment map obtained from the half-braiding (2.33) into (2.42), to obtain

the half-braiding back:



(2.43)

□

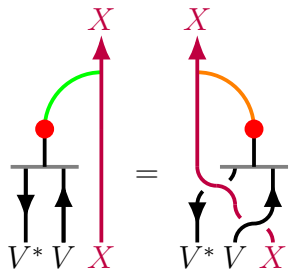
Proposition 2.15. Assume $\mathcal{C}$ is ribbon and contains the coend $\mathcal{L}$. Let (A, m_A, u_A) and (B, m_B, u_B) be algebras in $\mathcal{C}$ with quantum moment maps μ_A and μ_B respectively, and let (X, ρ_A, ρ_B) be an (A, B)-bimodule in $\mathcal{C}$. The functor

$$- \otimes_A X : \mathcal{C}_A \to \mathcal{C}_B \tag{2.44}$$

is a braided module functor if and only if

$$\begin{array}{ccccc} (V^* \otimes V) \otimes N & \xrightarrow{\mu_A \circ i_V \otimes N} & A \otimes N & \xrightarrow{\quad\rho_A\quad} & N \\ {\scriptstyle (V^* \otimes \sigma_{V,N}) \circ \alpha}\downarrow & & & & \uparrow{\scriptstyle \rho_B} \\ V^* \otimes (N \otimes V) & \xrightarrow[(\sigma^{-1}_{N,V^*} \otimes V) \circ \alpha]{} & (N \otimes V^*) \otimes V & \xrightarrow[(N \otimes \mu_B \circ i_V) \circ \alpha]{} & N \otimes B \end{array} \tag{2.45}$$

This condition is depicted graphically as

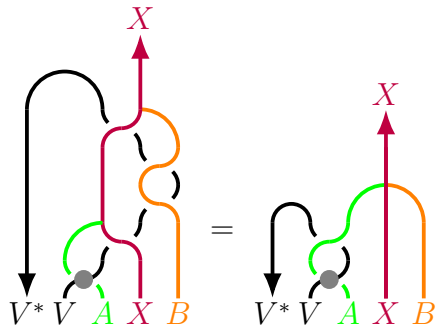


(2.46)

Proof. We show that the condition (2.23) implies the condition (2.46), which implies $- \otimes_A X$ is a braided $\mathcal{C}$-module functor. Proving the converse statement (that (2.46) implies (2.23)) is a similar computation, which we omit.

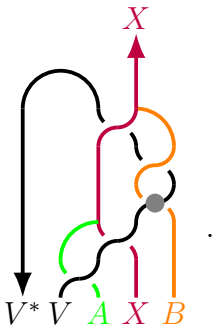
Invertibility of the braiding in $\mathcal{C}$ and the snake identities for the rigid structure

imply the left-hand side of (2.23) is equivalent to



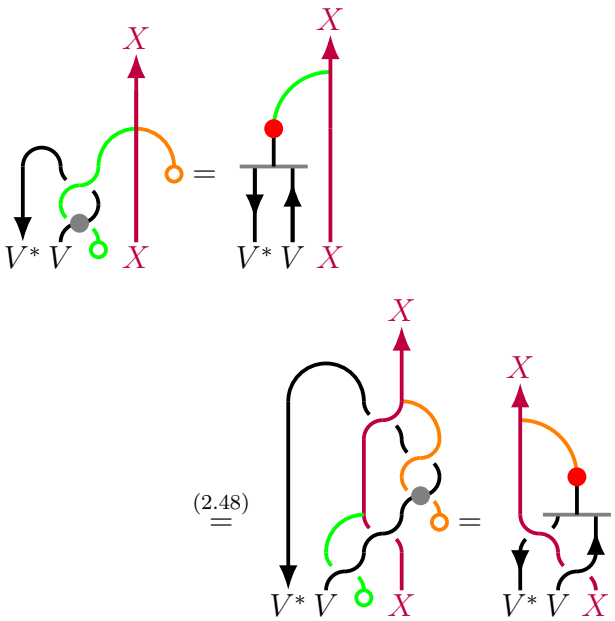
. (2.47)

Upon applying the same morphisms to the right-hand side of (2.23), we find



. (2.48)

Pre-composing with $V^* \otimes V \otimes u_A \otimes X \otimes u_B$ we obtain the identity (2.46):

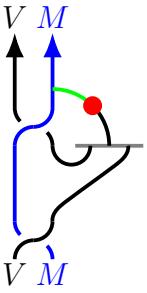


(2.49)

□

Combining Propositions 2.9 and 2.14, we arrive at the following result.

Corollary 2.16. [BZBJ18b, Cor. 4.11] Assume $\mathcal{C}$ is ribbon and contains the coend $\mathcal{L}$. Let (A, m, u) be an algebra in $\mathcal{C}$. There is bijective correspondence between module braidings for $\mathcal{C}_A$ and quantum moment maps $\mu : \mathcal{L} \to A$. In this correspondence, a module braiding is constructed from a quantum moment map via

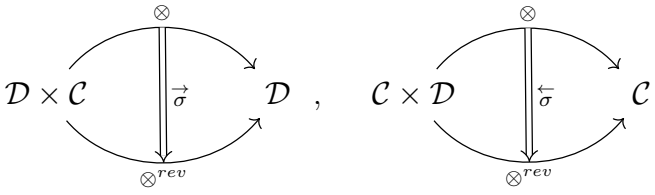
$e_{V,M} =$  . (2.50)

Remark 2.17. The pair (A, γ^A) is an object in the Drinfeld centre $Z(\mathcal{C})$ of $\mathcal{C}$. However, equipping the Drinfeld centre with the canonical monoidal structure which turns it into a braided monoidal category (see [EGNO15, Sec. 8.5]), the conditions (2.12) do not allow one to lift the algebra structure to an algebra structure in the Drinfeld centre. In [BZBJ18b, Cor. 4.6] (see also [JFR23, Def. 2.47]) it was observed that if $\mathcal{C}$ is ribbon, one can use the equivalence between $\mathcal{C}_{\mathcal{L}}$ and $Z(\mathcal{C})$ to define an alternative monoidal structure on $Z(\mathcal{C})$ such that a half-braided algebra in $\mathcal{C}$ lifts to an algebra in $Z(\mathcal{C})$. The category $\mathcal{C}_{\mathcal{L}}$ with this monoidal structure is sometimes called the *Harish-Chandra category* [GJS23, Sec. 2.5], see *loc. cit.* for details regarding its definition.

2.3 Small braidings

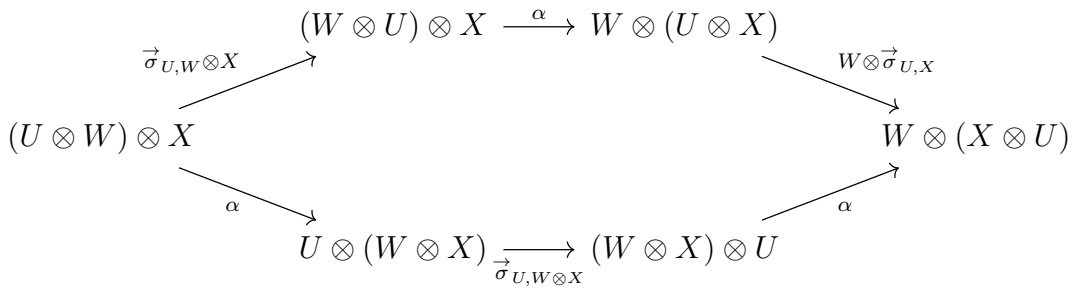
Let $(\mathcal{D}, \otimes, \mathbb{1}, \alpha, l, r)$ be a monoidal category, $\mathcal{C} \overset{\iota}{\hookrightarrow} \mathcal{D}$ a full monoidal subcategory, and $\otimes^{rev}$ the reversed tensor product: $U \otimes^{rev} V = V \otimes U$.

Definition 2.18. A *small braiding for* $\mathcal{C} \overset{\iota}{\hookrightarrow} \mathcal{D}$ is the data of two natural isomorphisms

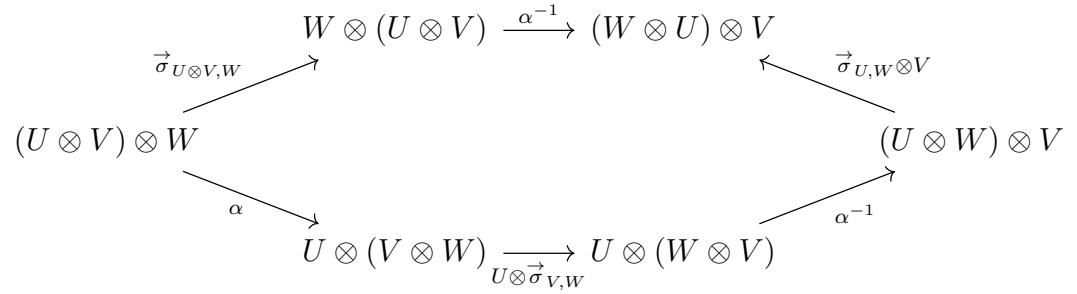


(2.51)

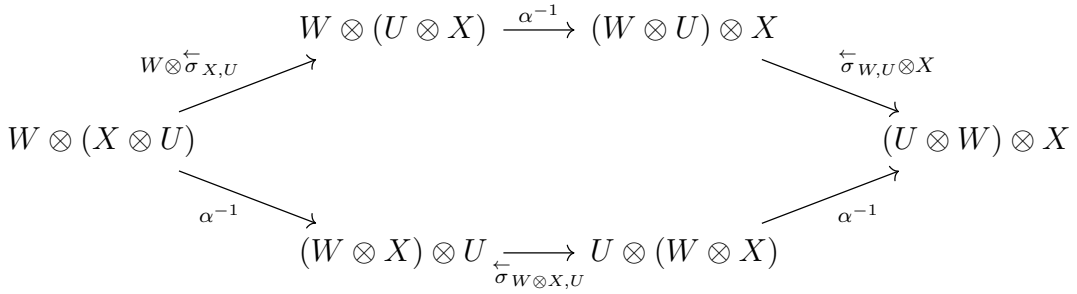
which satisfy the hexagon constraints,



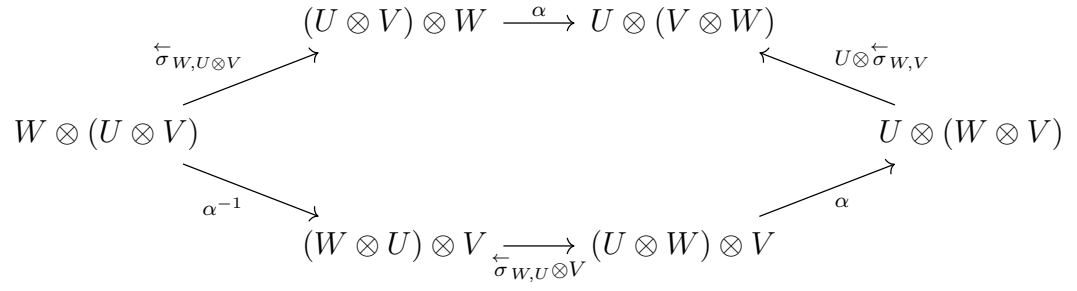
(2.52)



(2.53)



(2.54)



(2.55)

for all objects U, V in $\mathcal{D}$ and W, X in $\mathcal{C}$, and agree on $\mathcal{C}$. That is, for all objects U, V in $\mathcal{C}$, one has $\overrightarrow{\sigma}_{\iota U,V} = \overleftarrow{\sigma}_{U,\iota V}$.

We depict a small braiding similarly to a regular braiding by using a thicker

line to indicate the object which comes from $\mathcal{D}$:

$$\overrightarrow{\sigma}_{U,V} = \quad , \qquad \overleftarrow{\sigma}_{V,U} = \quad . \tag{2.56}$$



Lemma 2.19. A small braiding on $(\mathcal{C} \overset{\iota}{\hookrightarrow} \mathcal{D}, \overrightarrow{\sigma}, \overleftarrow{\sigma})$ endows $\mathcal{C}$ with the structure of a braided monoidal category, and $l_U \circ \overrightarrow{\sigma}_{U,\mathbb{1}} = r_U$ and $r_U \circ \overleftarrow{\sigma}_{\mathbb{1},U} = l_U$ for all $U \in \mathcal{D}$.

Proof. Since $\overrightarrow{\sigma}$ and $\overleftarrow{\sigma}$ agree on $\mathcal{C}$, their respective hexagons together define a braiding for $\mathcal{C}$. The unitality property $l_U \circ \overrightarrow{\sigma}_{U,\mathbb{1}} = r_U$ follows from setting $V = \mathbb{1}$ in (2.52) and inserting unitors in the interior as follows (c.f. [Ka95, Proposition XIII.1.2]):

$$\begin{array}{ccccccc}
 & & (\mathbb{1}\otimes U)\otimes W & \xrightarrow{\alpha} & \mathbb{1}\otimes(U\otimes W) & & \\
 & \overrightarrow{\sigma}_{U,\mathbb{1}}\otimes\mathrm{id}_W \nearrow & \downarrow{\scriptstyle l_U\otimes\mathrm{id}_W} & \swarrow{\scriptstyle l_{U\otimes W}} & & \searrow{\scriptstyle \mathbb{1}\otimes\overrightarrow{\sigma}_{U,W}} & \\
(U\otimes\mathbb{1})\otimes W & \xrightarrow[r_U\otimes\mathrm{id}_W]{} & U\otimes W & \xrightarrow[\overrightarrow{\sigma}_{U,W}]{} & W\otimes U & \xleftarrow[l_{W\otimes U}]{} & \mathbb{1}\otimes(W\otimes U) \\
 & \searrow{\scriptstyle \alpha} & {\scriptstyle \mathrm{id}_U\otimes l_W}\uparrow & & {\scriptstyle l_W\otimes\mathrm{id}_U}\uparrow & \nearrow{\scriptstyle \alpha} & \\
 & & U\otimes(\mathbb{1}\otimes W) & \xrightarrow[\overrightarrow{\sigma}_{U,\mathbb{1}\otimes W}]{} & (\mathbb{1}\otimes W)\otimes U & &
\end{array} \tag{2.57}$$

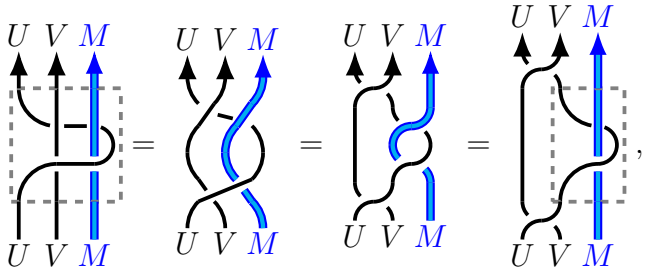
The square in the top half of the hexagon commutes via naturality of the unitor, and the square in the bottom half commutes from naturality of $\overrightarrow{\sigma}$. The two triangles adjacent to the top left triangle as well as the bottom right triangle commute from the definition of the unitor. Thus the top left triangle commutes, which gives us the result upon setting $W = \mathbb{1}$ and using the right unitor. The property $r_U \circ \overleftarrow{\sigma}_{\mathbb{1},U} = l_U$ can be proven by filling in the hexagon (2.54) similarly. □

In all sections that follow we denote with σ the underlying braiding in $\mathcal{C}$ coming from a small braiding $(\overrightarrow{\sigma}, \overleftarrow{\sigma})$ on $\mathcal{C} \overset{\iota}{\hookrightarrow} \mathcal{D}$. We also suppress coherence morphisms.

Proposition 2.20. Let $\mathcal{M}$ be a full subcategory of $\mathcal{D}$ such that $\iota V \otimes M$ is an object in $\mathcal{M}$ for all objects V in $\mathcal{C}$ and M in $\mathcal{M}$. Then $(\mathcal{M}, \overrightarrow{\sigma} \circ \overleftarrow{\sigma})$ is a braided left $\mathcal{C}$-module category with action functor $\iota \otimes -$.

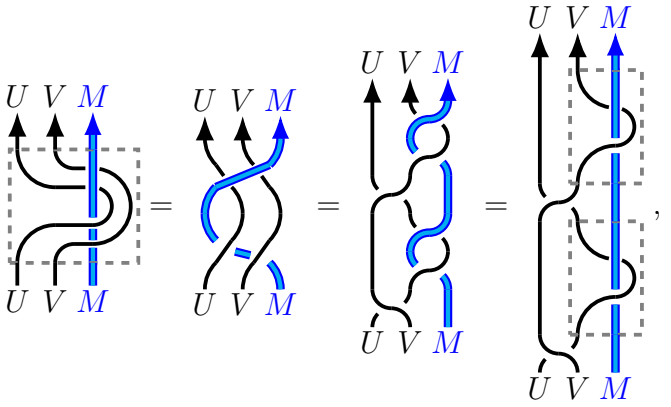
Proof. Let $\mathcal{M}$ be as above. The $\mathcal{C}$-module structure is given by the monoidal structure for $\mathcal{D}$. Let $U, V \in \mathcal{C}$ and $M \in \mathcal{M}$. The identities for $\overrightarrow{\sigma} \circ \overleftarrow{\sigma}$ we need to

verify are (2.6a) and (2.6b). The identity (2.6a) follows from



(2.58)

where in the first equality we inserted the definition of the module braiding $\overrightarrow{\sigma} \circ \overleftarrow{\sigma}$, and we used the hexagons in Definition 2.18 in the second equality. The identity (2.6b) follows from

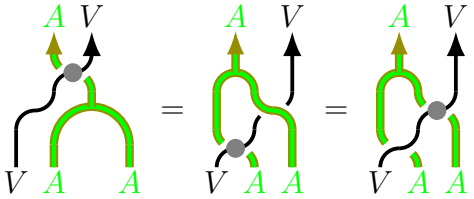


(2.59)

with the first equality being the definition of the module braiding and the second following from the hexagons and naturality of the small braidings □

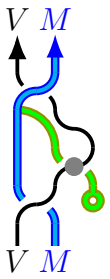
Assume $\mathcal{C}$ and $\mathcal{D}$ are abelian categories with right exact tensor product, and that $\iota : \mathcal{C} \hookrightarrow \mathcal{D}$ is right exact. Let (A, m, u) be an algebra in $\mathcal{D}$. By virtue of $\mathcal{C}$ being a monoidal subcategory of $\mathcal{D}$, the category $\mathcal{D}_A$ is a $\mathcal{C}$-module category. The following proposition adapts one direction of Proposition 2.9 in order to define a module braiding for $\mathcal{D}_A$ from a half-braiding for A. The proof is exactly the same as in Proposition 2.9.

Proposition 2.21. Assume $\mathcal{C}$ and $\mathcal{D}$ are abelian categories with a right exact tensor product. Let (A, m, u) be an algebra in $\mathcal{D}$. A half-braiding $\widehat{\gamma}^A : \iota\otimes A \Rightarrow A\otimes\iota$ satisfying



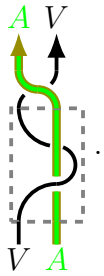
(2.60)

equips the $\mathcal{D}$-module category $\mathcal{D}_A$ with a module braiding given by

$e_{V,M} :=$ 

(2.61)

Remark 2.22. It is not clear to us if one can construct a half-braiding given a module braiding for $\mathcal{D}_A$: The inverse for the construction out of which one would wish to obtain a half-braiding is

$\widehat{\gamma}^A_V =$ 

.
(2.62)

The first equality in (2.60) is implied from each $e_{V,A}$ being a morphism in $\mathcal{D}_A$ and the fact that the small braidings are natural. We do not know how to prove the second equality from the basic properties of module braidings. That being said, we do not have an example of a module braiding for $\mathcal{D}_A$ that does not give A a half-braiding.

As in the previous subsection, we call the pair consisting of an algebra $(A, m, u) \in \mathcal{D}$ and a half-braiding $\widehat{\gamma}^A$ satisfying (2.60) a half-braided algebra in $\mathcal{D}$.

Proposition 2.23. Upon replacing braidings with the appropriate half-braidings, Proposition 2.12 generalises to the case where A and B are half-braided algebras in $\mathcal{D}$.

Proof. Proving that the half-braided bimodule condition (2.23) is necessary and sufficient is done exactly as in Proposition 2.12. The only thing we require in addition is that for $A, B \in \mathcal{C}$, $M \in \mathcal{C}_A$ and X an (A, B)-bimodule in $\mathcal{C}$, the ιB-modules $\iota(M \otimes_A X)$ and $\iota M \otimes_{\iota A} \iota X$ should agree. Namely, we wish for ι to preserve coequalisers. This follows from our assumption that ι is right exact. □

Assume in addition to $\mathcal{C}$ and $\mathcal{D}$ being abelian and ι being right exact, that $\mathcal{C}$ is a ribbon category and that ι is left exact. Assume that the coend of the functor

$\iota((-)^* \otimes =) : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{D}$, denoted

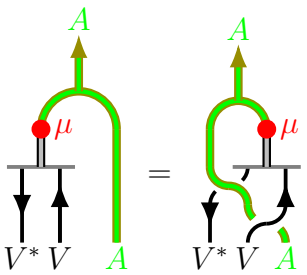
$$\mathcal{L} := \int^{V \in \mathcal{C}} \iota(V^* \otimes V) \,, \tag{2.63}$$

is contained in $\mathcal{D}$. In order to make sense of the algebra structure (2.37a),(2.37b) for $\mathcal{L}$ we require the Fubini theorem for coends,

$$\int^{V \in \mathcal{C}} \int^{U \in \mathcal{C}} \iota(U^* \otimes U) \otimes \iota(V^* \otimes V) \cong \int^{U \in \mathcal{C}} \int^{V \in \mathcal{C}} \iota(U^* \otimes U) \otimes \iota(V^* \otimes V) \,, \tag{2.64}$$

to hold. Since we ι assume is left exact, this follows from [Ly96, Thm. B.2]. As we did in Proposition 2.21, we obtain one direction of the results in Proposition 2.14 and Corollary 2.16.

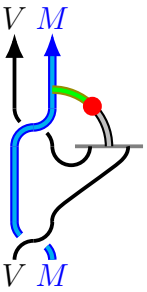
Proposition 2.24. Let $\mathcal{C}$ be ribbon, $\mathcal{D}$ contain the coend $\mathcal{L}$, and $\iota : \mathcal{C} \hookrightarrow \mathcal{D}$ be exact. Let (A, m, u) be an algebra in $\mathcal{D}$. Let $\mu : \mathcal{L} \to A$ be an algebra homomorphism in $\mathcal{D}$ satisfying



(2.65)

for all $V \in \mathcal{C}$. The morphisms

$e_{V,M} =$



(2.66)

form a module braiding for $\mathcal{D}_A$.

Remark 2.25. Let $\mathcal{C}$ be ribbon, $\mathcal{D}$ contain the coend $\mathcal{L}$, and $\iota : \mathcal{C} \hookrightarrow \mathcal{D}$ be exact. Upon replacing braidings with the appropriate small braidings, Proposition 2.15 holds for algebras $A, B \in \mathcal{D}$.

3 Braided module categories for quantum groups

In this section we construct examples of braided module categories and half-braided algebras/quantum moment maps coming from quantum groups. First we discuss small braidings associated to quantum groups at generic quantum parameter and introduce several concrete examples. We then change settings, and discuss small braidings coming from the $\hbar$-adic quantum group $U_\hbar\mathfrak{g}$. In the $\hbar$-adic setting we apply the machinery of Appendix A and the previous section to construct quantum moment maps from internal-hom algebras.

Throughout this section, fix a semisimple Lie algebra $\mathfrak{g}$ of rank n. Let $\{\alpha_i\}$ be the simple roots of $\mathfrak{g}$, $\{H_i\}$ the simple coroots and P the weight lattice. Let $(-,=)$ be the standard symmetric bilinear form on $\mathfrak{h}^*$ defined via the Killing form on $\mathfrak{g}$, and $d_i = (\alpha_i, \alpha_i)$.

3.1 Small braidings from unrolled quantum groups

We work over $\mathbb{C}$ and with quantum parameter $q \in \mathbb{C}^\times$, for which we fix a logarithm $\log q \in \mathbb{C}$. We say q is *generic* if it is not a root of unity, that is, $\log q \notin \pi\sqrt{-1}\mathbb{Q}$.

We denote $e^{a \log q}$ by q^a, where a can be a number or an operator. We denote with $U_q^H\mathfrak{g}$ the unrolled quantum group based on $\mathfrak{g}$, which was originally defined for $\mathfrak{g} = \mathfrak{sl}_2$ in [GPT07] and for an arbitrary semisimple Lie algebra in [GP10]. Concretely, we use the definition of $U_q^H\mathfrak{g}$ from [Ru19, Def. 3.1], which allows for generic quantum parameter.

Let us briefly recall a few basic facts and notation. Denote $U_q\mathfrak{h}$, $U_q\mathfrak{n}_+$ and $U_q\mathfrak{n}_-$ the subalgebras of $U_q^H\mathfrak{g}$ generated by $\{K_i^{\pm 1}\}_{i=1}^n$, $\{E_i\}_{i=1}^n$ and $\{F_i\}_{i=1}^n$, respectively, and denote the generators of $U\mathfrak{h}$ with $\{H_i\}_{i=1}^n$. The translation between the notation of [Ru19] and our notation is $K_i = K_i$, $K_{-i} = K_i^{-1}$, $X_i = E_i$ and $X_{-i} = F_i$. The generators coming from $U_q\mathfrak{n}_\pm$ act nilpotently in finite-dimensional representations. Let $U_q\mathfrak{g}$ be the subalgebra of $U_q^H\mathfrak{g}$ generated by $U_q\mathfrak{h}$ and $U_q\mathfrak{n}_\pm$. This agrees with the definition of $U_q\mathfrak{g}$ given in [KS97, Sec. 6.1.2].

Recall that a $U_q^H\mathfrak{g}$-module V is a *weight module* if there exist vectors $\lambda \in \mathfrak{h}^*$ such that V decomposes as [GP10, Sec. 5.3]

$$V \cong \bigoplus_{\lambda\in\mathfrak{h}^*} V_\lambda \ , \quad V_\lambda = \{v : H_i \cdot v = \lambda(H_i)v \ , \quad K_i \cdot v = q^{H_i} \cdot v\} \ . \tag{3.1}$$

We do not demand the V_λ to be finite-dimensional.

For applications in skein theory, we are interested in defining a small braiding for the category of so-called finite-dimensional type 1 modules for $U_q\mathfrak{g}$. Recall that a $U_q\mathfrak{g}$-module is of type 1 if there exist $\lambda \in P$ such that [CP95, pp. 314] (see also

[Ch94, pp. 4])

$$V \cong \bigoplus_{\lambda \in P} V_\lambda \ , \quad V_\lambda = \{v : K_i \cdot v = q^{\lambda(H_i)} v\} \ . \tag{3.2}$$

Let $\mathcal{D}^H$ be the full monoidal subcategory of $U_q^H\mathfrak{g}$-modules consisting of weight modules, $\mathcal{C}^H$ the full monoidal subcategory of $\mathcal{D}^H$ consisting of finite-dimensional modules, and $\mathcal{C}$ be the category of finite-dimensional $U_q\mathfrak{g}$-modules of type 1.

It is well-known that finite-dimensional $U_q\mathfrak{g}$-modules of type 1 form a braided monoidal category, see for example [KS97, Prop. 8.19] (but there is no braiding for all finite-dimensional $U_q\mathfrak{g}$-modules, see [WW26]). The braiding is constructed in [KS97, Sec. 8.3.3] by defining an endomorphism of the tensor product of $U_q\mathfrak{g}$-modules which implements the action of the universal R-matrix for the $\hbar$-adic quantum group with the formal parameter being specialised to a complex number. Replacing the formal parameter with a complex number does not recover an element of the tensor square of $U_q\mathfrak{g}$, so one cannot obtain a universal R-matrix for $U_q\mathfrak{g}$ this way. To get around this, one defines an element R of a completion $\widehat{U_q\mathfrak{g} \otimes U_q\mathfrak{g}}$ of the tensor square of $U_q\mathfrak{g}$ [KS97, Thm. 8.18], which contains infinite sums over powers of $U_q\mathfrak{n}_- \otimes U_q\mathfrak{n}_+ \subset \widehat{U_q\mathfrak{g} \otimes U_q\mathfrak{g}}$ generators, and an endomorphism Φ of the tensor product of $U, V \in \mathcal{C}$ which acts as $u \otimes v \mapsto q^{(\lambda,\mu)} u \otimes v$ for $u \in U_\lambda$ and $v \in V_\mu$ [KS97, Prop. 8.19], such that

$$u \otimes v \mapsto \Phi R \cdot (u \otimes v) \tag{3.3}$$

implements the action of the $\hbar$-adic R-matrix with the formal variable specialised to a complex number. Composing this with the swap operation of vector spaces then implements a braiding.

We now outline how this braiding can be extended to finite-dimensional $U_q^H\mathfrak{g}$-weight modules. Using the duality between $\mathfrak{h}^*$ and $\mathfrak{h}$ the endomorphism $u \otimes v \mapsto q^{(\lambda,\mu)} u \otimes v$ can be realised in terms of the generators of $U\mathfrak{h} \subset U_q^H\mathfrak{g}$ [GP10, Lem. 40],

$$u \otimes v \mapsto q^{\sum_{i,j=1}^n d_i (A^{-1})_{ij} H_i \otimes H_j} \cdot u \otimes v \ . \tag{3.4}$$

This endomorphism is well-defined on the tensor product of any two weight modules for $U_q^H\mathfrak{g}$, regardless of whether or not their weight supports are integral. Since $U_q\mathfrak{g}$ is a subalgebra of $U_q^H\mathfrak{g}$, one can ask whether the morphisms (3.3), where Φ is replaced by the endomorphism (3.4), define a braiding between finite-dimensional $U_q^H\mathfrak{g}$-weight modules. Because

$$q^{\sum_{i,j=1}^n d_i (A^{-1})_{ij} H_i \otimes H_j} \Delta(H_k) = \Delta(H_k) q^{\sum_{i,j=1}^n d_i (A^{-1})_{ij} H_i \otimes H_j} \tag{3.5}$$

for all $k = 1, \ldots, n$, it suffices to check $R\Delta(H_i) = \Delta^{\mathrm{op}}(H_i)R$. A straightforward computation shows this to be true.

Remark 3.1. The truncated R-matrices of [GP10, Ru19] can only be used to implement a braiding when q is a root of unity and one restricts to a certain subcategory of $\mathcal{C}^H$. Alternatively, one can take the quotient of $U_q^H\mathfrak{g}$ by ideals generated by certain powers of the generators coming from $U_q\mathfrak{n}_\pm$ to define the restricted quantum group [GPT07, CGP14], the modules over which admit a braiding in terms of a truncated R-matrix.

The obstruction to braiding $U_q^H\mathfrak{g}$-modules of arbitrary dimension is that the sum over powers of the $U_q\mathfrak{n}_- \otimes U_q\mathfrak{n}_+$ generators in the R-matrix will not truncate when acting on a tensor product of modules which are not locally finite as $U_q\mathfrak{n}_+$- or $U_q\mathfrak{n}_-$-modules, see [KS97, Thm. 8.18] for the explicit form of the R-matrix. The torsion-free modules of [FHW15] are an example of such modules. However, the sum will truncate if at least one of the modules is finite-dimensional, so one can braid a finite-dimensional module with an infinite-dimensional module. We then arrive at our first example of a small braiding:

Proposition 3.2. For all $U, V \in \mathcal{D}^H$ define $s_{U,V}$ to be the swap operation on tensor products of vector spaces: $s_{U,V}(u \otimes v) := v \otimes u$ for all $u \in U$ and $v \in V$. If q is generic, then the morphisms

$$\overrightarrow{\sigma}_{U,V}(u \otimes v) = s_{U,V} q^{\sum_{i,j=1}^n d_i (A^{-1})_{ij} H_i \otimes H_j} R \cdot (u \otimes v) \tag{3.6a}$$

$$\overleftarrow{\sigma}_{V,U}(v \otimes u) = s_{V,U} q^{\sum_{i,j=1}^n d_i (A^{-1})_{ij} H_i \otimes H_j} R \cdot (v \otimes u)\ , \tag{3.6b}$$

where R is the R-matrix for $U_q\mathfrak{g}$, defines a small braiding on $\mathcal{C}^H \hookrightarrow \mathcal{D}^H$.

The following proposition allows us to define a small braiding between finite-dimensional $U_q\mathfrak{g}$-modules of type 1 and $U_q^H\mathfrak{g}$-weight modules.

Proposition 3.3. . Let q be generic and $\mathcal{C}$ be the category of finite-dimensional $U_q\mathfrak{g}$-modules of type 1. There is a full braided monoidal inclusion of categories $\iota : \mathcal{C} \hookrightarrow \mathcal{C}^H$.

Proof. If q is generic the decomposition (3.2) of $V \in \mathcal{C}$ is unique as each $\lambda(H_i) \in \mathbb{Z}$ when $\lambda \in P$. Define the functor $\iota : \mathcal{C} \to \mathcal{C}^H$ to leave the underlying vector spaces of the objects and linear maps of the morphisms the same, and define an action of the generators of $U\mathfrak{h}$ to be $H_i \cdot v = \lambda(H_i)v$ for $v \in V_\lambda$. The uniqueness of the decomposition (3.2) implies that linear maps which intertwine the $U_q\mathfrak{h}$ generators also intertwine the generators of $U\mathfrak{h}$, so ι is a well-defined functor. Furthermore the uniqueness of the decomposition implies such maps are in bijection, so ι is fully faithful. Let us now show ι is monoidal. Let $u \in U_\mu$ and $v \in V_\lambda$ for $U, V \in \mathcal{C}$. Consider $u \otimes v \in (U \otimes V)_{\mu+\lambda}$, on which the action of $K_i \in U_q\mathfrak{h}$ is given by

$$K_i \cdot (u \otimes v) = q^{\mu(H_i)+\lambda(H_i)} u \otimes v = q^{(\mu+\lambda)(H_i)} u \otimes v\ . \tag{3.7}$$

The action of $H_i \in U\mathfrak{h}$ on $u \otimes v$ is

$$H_i \cdot (u \otimes v) = H_i \cdot u \otimes v + u \otimes H_i \cdot v = (\mu + \lambda)(H_i)u \otimes v \ , \tag{3.8}$$

as desired. We see that this inclusion is braided because identification of the endomorphism Φ in (3.3) with $q^{\sum_{i,j=1}^{n} d_i (A^{-1})_{ij} H_i \otimes H_j}$ [GP10, Lem. 40] □

The previous two propositions now imply the main theorem of this section:

Theorem 3.4. Let $\mathcal{C}$ be the category of finite-dimensional $U_q\mathfrak{g}$-modules of type 1, $\mathcal{D}^H$ be the category $U_q^H\mathfrak{g}$-weight modules, and $\mathcal{C}^H$ the full subcategory of $\mathcal{D}^H$ of finite-dimensional modules. For generic quantum parameter q, there is a small braiding on the monoidal inclusion $\mathcal{C} \overset{\iota}{\hookrightarrow} \mathcal{C}^H \hookrightarrow \mathcal{D}^H$ given by

$$\overrightarrow{\sigma}_{U,V}(u \otimes v) = s_{U,V} q^{\sum_{i,j=1}^{n} d_i (A^{-1})_{ij} H_i \otimes H_j} R \cdot (u \otimes v) \tag{3.9a}$$

$$\overleftarrow{\sigma}_{V,U}(v \otimes u) = s_{V,U} q^{\sum_{i,j=1}^{n} d_i (A^{-1})_{ij} H_i \otimes H_j} R \cdot (v \otimes u) \ , \tag{3.9b}$$

where R is the R-matrix for $U_q\mathfrak{g}$.

3.2 Examples for generic quantum parameter

We now study subcategories of $\mathcal{D}^H$ closed under tensoring with objects in $\mathcal{C}$ in order to construct examples of braided module categories from the small braiding on $\mathcal{C} \hookrightarrow \mathcal{D}^H$ in Theorem 3.4.

In the examples that follow, the module braiding is given by the monodromy $\overrightarrow{\sigma} \circ \overleftarrow{\sigma}$, as per the perscription in Lemma 2.20. We understand the following example to be folklore and have not seen it not written down anywhere in the literature. It can be viewed as an unrolled version of the quantum category $\mathcal{O}$ of [AM14].

Example 3.5. Define $\mathcal{O}_q^H$ to be the full subcategory of $\mathcal{D}^H$ consisting of modules which are finitely generated and in which $U_q^H\mathfrak{n}_+$ acts nilpotently. $\mathcal{O}_q^H$ is closed under tensoring with $\mathcal{C}$ and thus a braided $\mathcal{C}$-module category.

For each pair $\lambda, \mu \in \mathfrak{h}^*$, say $\lambda \sim \mu$ if $\lambda - \mu$ is integral. Let $W \in \mathcal{C}$ and $V \in \mathcal{D}^H$. Because the weight support of W is integral, given a weight λ in the support of V, any weight μ in the support of $W \otimes V$ is such that $\mu \sim \lambda$. We then arrive at the following example of a subcategory of $\mathcal{D}^H$ closed under tensoring with objects in $\mathcal{C}$.

Example 3.6. For each $\Lambda \in \mathfrak{h}^*/\sim$, the full subcategory $\mathcal{D}_\Lambda^H$ of $\mathcal{D}^H$ consisting of modules with weight support in Λ is closed under tensoring with objects in $\mathcal{C}$ and thus a braided $\mathcal{C}$-module category.

Let $[0] \in \mathfrak{h}^*/\sim$ be the class containing integral weights. Weights which are not in $[0]$ are referred to as generic. Given $\lambda \in \mathfrak{h}^*$ one can define a Verma module $V(\lambda)$ for the unrolled quantum group analogously to the construction of a Verma module for $U\mathfrak{g}$ [GP10, Sec. 5.5]. As in the classical case, $V(\lambda)$ is irreducible if λ is not dominant integral [ES00, Prop. 2.1]. Using this, we define module subcategories of $\mathcal{D}_\Lambda^H$ which contain only direct sums of irreducible Verma modules. First we show how the tensor product of an irreducible Verma module with a finite-dimensional module decomposes as a direct sum of irreducible Verma modules.

Proposition 3.7. Let $\lambda \in \mathfrak{h}^*$ be generic and $V(\lambda)$ be a Verma module for $U_q^H\mathfrak{g}$. The tensor product $W \otimes V(\lambda)$ decomposes as

$$W \otimes V(\lambda) \cong \bigoplus_{\mu\in\mathfrak{h}^*} V(\lambda+\mu)^{\oplus \dim(W_\mu)} \ . \tag{3.10}$$

for all finite-dimensional weight modules W.

Proof. Given $\lambda, \mu \in \mathfrak{h}^*$ we denote with $v_\lambda \in V(\lambda)$ the highest weight vector. It is shown in [ES00, Prop. 2.2] that for $\lambda \in \mathfrak{h}^*$ generic, there is for every $w \in W_\mu$ a unique intertwiner $\Phi^w_{\lambda+\mu} : V(\lambda+\mu) \to W \otimes V(\lambda)$ such that

$$\Phi^w_{\lambda+\mu}(v_{\lambda+\mu}) = w \otimes v_\lambda + u \ , \tag{3.11}$$

where $u \in \bigoplus_{\nu<\lambda} W \otimes V(\lambda)_\nu$. Combining these, we obtain an intertwiner

$$\eta^\lambda_W : \bigoplus_{\mu\in\mathfrak{h}^*} W_\mu \otimes V(\lambda+\mu) \to W \otimes V(\lambda) \quad , \quad w \otimes v \mapsto \Phi^w_{\lambda+\mu}(v) \ , \tag{3.12}$$

where the action of $U_q^H\mathfrak{g}$ on $W_\mu \otimes V(\lambda+\mu)$ is only on the second factor, see [Ko00, Eq. 6.10]. As a $U_q^H\mathfrak{g}$-module, the domain of η^λ_W is isomorphic to $\bigoplus_{\mu\in\mathfrak{h}^*} V(\lambda+\mu)^{\oplus\dim(W_\mu)}$. To establish (3.10), it thus remains to show that η^λ_W is an isomorphism.

We begin with injectivity of η^λ_W. First note that by construction the linear map

$$W = \bigoplus_{\mu\in\mathfrak{h}^*} W_\mu \to W \otimes V(\lambda) \ , \quad w \mapsto \Phi^w_{\lambda+\mu}(v_{\lambda+\mu}) \quad (\text{for } w \in W_\mu) \tag{3.13}$$

is injective. Because M is finite-dimensional, its weight support is integral, hence each $V(\lambda+\mu)$ is irreducible. It then follows that if the kernel of η^λ_W is non-zero, it must also contain an element of the form $w \otimes v_{\lambda+\mu}$, which by the above observation cannot happen.

Now we show η^λ_W is surjective. The restriction of η^λ_W to each weight space is also an injective map. Because the weight spaces of the left- and right-hand side of

(3.10) are finite-dimensional, if they have equal dimension, η_W^λ is surjective. This in turn follows from

$$\begin{aligned}\dim(W \otimes V(\lambda))_\eta &= \sum_\mu \dim(W_\mu)\dim(V(\lambda)_{\eta-\mu}) = \sum_\mu \dim(W_\mu)\dim(V(\lambda+\mu)_\eta) \\ &= \dim\Big(\bigoplus_\mu W_\mu \otimes V(\lambda+\mu)\Big)_\eta \,. \end{aligned} \tag{3.14}$$

□

Now we can define an example containing only direct sums of irreducible Verma modules.

Example 3.8. For each $\Lambda \in (\mathfrak{h}^*/\sim) \setminus [0]$ the full subcategory $\mathcal{M}_\Lambda^H$ of $\mathcal{D}_\Lambda^H$ of modules that are isomorphic to a finite direct sum of Verma modules of highest weight $\lambda \in \Lambda$ is closed under tensoring with $\mathcal{C}$, and thus a braided $\mathcal{C}$-module category.

In the case $\mathfrak{g} = \mathfrak{sl}_2$, facets of the categories $\mathcal{M}_\Lambda^H$ have appeared in the literature before. Fix the pivotal structure for $\mathcal{C}$ which recovers the Kauffman bracket skein relations (see [Ti10] for more details) and let V be the fundamental representation of $U_q\mathfrak{sl}_2$. In this setting the Temperley-Lieb algebra on n strands is isomorphic to the endomorphism space of $V^{\otimes n} \in \mathcal{C}$. Furthermore, the hom-space from n to m in the Temperley-Lieb category is isomorphic to $\mathcal{C}(V^{\otimes m}, V^{\otimes n})$, and the Karoubi completion of the Temperley-Lieb category is $\mathcal{C}$. From the results of [ILZ19], the blob algebra of [MS94] on n strands is isomorphic to the endomorphism algebra of $V^{\otimes n} \triangleright V(\lambda)$ in $\mathcal{M}_\Lambda^H$. In fact, the hom-spaces of the blob category defined in [LNV21, Sec. 2.2] are given by hom-spaces of $\mathcal{M}_\Lambda^H$. We also deduce from this observation that the blob category is a braided module category over the Temperley-Lieb category. See also [CGS22] for the relation between endomorphism spaces of $\mathcal{M}_\Lambda^H$, the blob algebra, and boundary conditions of the XXZ spin chain.

3.3 Quantum moment maps from internal-homs

We now construct examples of quantum moment maps, namely those which recover the quantisations of [Mu07]. In order to do this we change settings and work with the $\hbar$-adic Hopf algebra $U_\hbar\mathfrak{g}$ over the ring $\mathbb{C}[[\hbar]]$ of power series in a formal parameter $\hbar$, see [KS97, Sec. 6.1.3] for details regarding the definition. First we introduce some terminology which will let us define categories of $U_\hbar\mathfrak{g}$-modules from which we can obtain quantum moment maps via small braidings.

Let W be a complex vector space. Define $W[[\hbar]]$ to be the $\mathbb{C}[[\hbar]]$-module containing all formal series $\sum_{n\geq 0} w_n\hbar^n$, where $(w_0, w_1, \dots)$ is a sequence of elements in

W (see [Ka95, Eq. (XVI.1.2)] for the $\mathbb{C}[[\hbar]]$-module structure). A $U_\hbar\mathfrak{g}$-module V is called *topologically free* if it is isomorphic as a $\mathbb{C}[[\hbar]]$-module to $W[[\hbar]]$, where W is a $U\mathfrak{g}$-module over $\mathbb{C}$ [Ka95, Sec. XVI.2]. A topologically free module $V \cong W[[\hbar]]$ is said to be of *finite-rank* if W is a finite-dimensional $U\mathfrak{g}$-module [Ka95, Sec. XVII.3].

Let $I_\hbar$ be the left principal ideal in $\mathbb{C}[[\hbar]]$ generated by $\hbar$. Let $\mathcal{D}$ be the category of derived $I_\hbar$-adic complete $U_\hbar\mathfrak{g}$-modules, see Appendix B for details regarding the definition. The completed tensor product (see Definition B.2) of topologically free modules $X[[\hbar]]$ and $Y[[\hbar]]$ is given by $X[[\hbar]]\widehat{\otimes}Y[[\hbar]] \cong (X \otimes Y)[[\hbar]]$ as $\mathbb{C}[[\hbar]]$-modules, so topologically free modules form a monoidal subcategory of $U_\hbar\mathfrak{g}$-modules [Ka95, Prop. XVI.3.2]. It follows from [Ka95, Prop. XVI.2.4] and [PW22, Prop. 3.7] that every topologically free $U_\hbar\mathfrak{g}$-module is derived $I_\hbar$-adic complete, making topologically free $U_\hbar\mathfrak{g}$-modules a full monoidal subcategory of $\mathcal{D}$. The Hopf algbra $U_\hbar\mathfrak{g}$ is itself topologically free, and isomorphic to $U\mathfrak{g}[[\hbar]]$ as an algebra [Ka95, Thm. XVIII.4.1], hence $\mathcal{D}$ is abelian, see Proposition B.4. We can apply the machinery of Section 2.3 and Appendix A in order to construct quantum moment maps and inclusions of braided module categories.

Remark 3.9. The category of topologically free $U_\hbar\mathfrak{g}$-modules, however this is not an abelian category. Consider the endormorphism of $\mathbb{C}[[\hbar]]$ given by $f : r \mapsto r\hbar$. f is injective, hence a monomorphism, and it follows from $\mathbb{C}[[\hbar]]$-linearity that it is an epimorphism. However, f is not surjective and thus not an isomorphism, which implies topologically free modules do not form an abelian category.

In what follows we use the derived completed tensor product $\widehat{\otimes}_0$, which coincides with the completed tensor product $\widehat{\otimes}$ when taking in topologically free modules. The Hopf algebra $U_\hbar\mathfrak{g}$ is a quasitriangular $\hbar$-adic Hopf algebra [KS97, Thm 8.17], meaning the R-matrix $\mathcal{R} = \mathcal{R}_1\widehat{\otimes}_0\mathcal{R}_2$ for $U_\hbar\mathfrak{g}$ is an element of $U_\hbar\mathfrak{g}\widehat{\otimes}_0 U_\hbar\mathfrak{g}$. Hence acting with $\mathcal{R}$ is well-defined on any tensor product of derived $I_\hbar$-adic complete $U_\hbar\mathfrak{g}$-modules, not just the finite-dimensional ones. We then have that the category $\mathcal{D}$ of derived $I_\hbar$-adic complete $U_\hbar\mathfrak{g}$-modules is a braided monoidal category. From [Ka95, Exercise XVI.2] we also see that $\mathcal{D}$ is closed, with the internal-hom objects $\underline{\mathcal{D}}(U, V)$ being given by $\mathrm{Hom}_{\mathbb{C}[[\hbar]]}(U, V)$. The action of $U_\hbar\mathfrak{g}$ on $\underline{\mathcal{D}}(U, V)$ is

$$(x \cdot f)(-) = x_{(1)} f(S(x_{(2)}) \cdot -) \tag{3.15}$$

for all $x \in U_\hbar\mathfrak{g}$ and $f \in \underline{\mathcal{D}}(U, V)$.

Now we fix our setting: Let $\mathcal{C}$ be the category of finite-rank topologically free $U_\hbar\mathfrak{g}$-modules and $\mathcal{D}$ be the category of derived $I_\hbar$-adic complete $U_\hbar\mathfrak{g}$-modules. When we write the tensor product $\otimes$ we always mean $\widehat{\otimes}_0$. The small braiding on $\mathcal{C} \hookrightarrow \mathcal{D}$ is simply given by the braiding on $\mathcal{D}$. We indicate successive applications of $\mathcal{R}$ with a prime superscript. For example, the monodromy $\overrightarrow{\sigma} \circ \overleftarrow{\sigma}$ is given by

$$\overrightarrow{\sigma}_{V,U} \circ \overleftarrow{\sigma}_{U,V}(u \otimes v) = \mathcal{R}_2'\mathcal{R}_1 u \otimes \mathcal{R}_1'\mathcal{R}_2 v \ , \tag{3.16}$$

for all $U \in \mathcal{C}$, $V \in \mathcal{D}$ and $u \in U$, $v \in V$.

Fix a full subcategory $\mathcal{M} \subset \mathcal{D}$ which is closed under tensoring with objects in $\mathcal{C}$. We also use the fact that $\mathcal{D}$ is a closed category, and $\mathcal{C}$ is rigid [Ka95, Sec. XVII.3], which allows us to apply the results of Appendix A. Denote by $\underline{\mathcal{D}}(V)$ the internal endomorphism algebra $(\underline{\mathcal{D}}(V,V), m^V_{V,V}, u_V)$ of Corollary A.4.

Proposition 3.10. For all $V \in \mathcal{D}$, the natural isomorphism $\gamma^{\underline{\mathcal{D}}(V)}_{-} : - \otimes \underline{\mathcal{D}}(V) \Rightarrow \underline{\mathcal{D}}(V) \otimes -$ defined via

$$\begin{array}{ccccc} U \otimes \underline{\mathcal{D}}(V) & & \xrightarrow{\quad\Phi_{U,V}\quad} & & \underline{\mathcal{D}}(V, U \otimes V) \\ \big\downarrow{\scriptstyle \gamma^{\underline{\mathcal{D}}(V)}_{U}} & & & & \big\downarrow{\scriptstyle \underline{\mathcal{D}}(V, \vec{\sigma}_{V,U} \overleftarrow{\sigma}_{U,V})} \\ \underline{\mathcal{D}}(V) \otimes U & \xrightarrow{\vec{\sigma}_{U,\underline{\mathcal{D}}(V)}} & U \otimes \underline{\mathcal{D}}(V) & \xrightarrow{\Phi_{U,V}} & \underline{\mathcal{D}}(V, U \otimes V) \end{array} \tag{3.17}$$

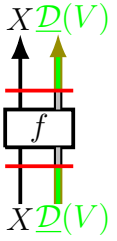
is a half-braiding for $\underline{\mathcal{D}}(V)$. Let $M \in \mathcal{M}$ and consider $\mathcal{M}$ and $\mathcal{D}_{\underline{\mathcal{D}}(M)}$ as braided module categories, with module braiding coming from the small braiding and half-braiding respectively. The functor $\underline{\mathcal{D}}(M, -) : \mathcal{M} \to \mathcal{D}_{\underline{\mathcal{D}}(M)}$ is a braided $\mathcal{C}$-module functor.

Proof. Naturality of the half-braiding follows from naturality of the coherence morphisms and the small braidings. It remains to show that the identities (2.60) hold.

Let $X, Y \in \mathcal{C}$. Conjugating the image of a morphism $f : X \otimes V \to Y \otimes V$ under $\underline{\mathcal{D}}(V, -)$ with the $\mathcal{C}$-module functor structure Φ for $\underline{\mathcal{D}}(V, -)$ defines a morphism

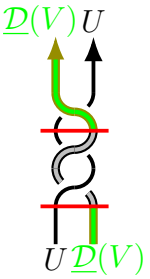
$$\underline{f} := \Phi_{Y,V} \underline{\mathcal{D}}(V, f) \Phi^{-1}_{X,V} : X \otimes \underline{\mathcal{D}}(V) \to Y \otimes \underline{\mathcal{D}}(V) \,. \tag{3.18}$$

We depict this as follows.

$\underline{f} =$ 

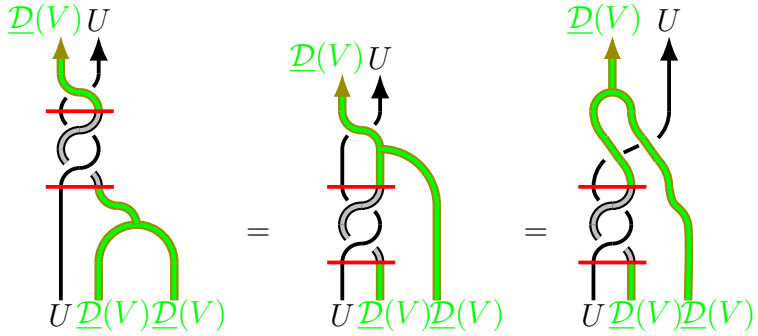
(3.19)

The candidate (3.17) for the half-braiding is then depicted

$\gamma^{\underline{\mathcal{D}}(V)}_{U} =$ 

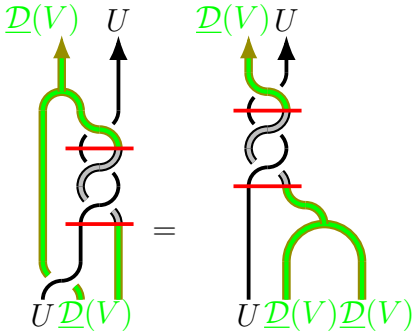
(3.20)

The first equality in (2.60) then follows from the fact that $\underline{f}$ is a morphism in $\mathcal{C}_{\underline{\mathcal{D}}(V)}$, and the naturality of the small braidings:



(3.21)

The second equality in (2.60) is depicted



(3.22)

We show that this holds via a direct calculation.

We begin by evaluating the right-hand side of (3.22) on a vector in $U \otimes (\underline{\mathcal{D}}(V) \otimes \underline{\mathcal{D}}(V))$. First we show that the morphism

$$\begin{aligned} \phi_{U,V} : U \otimes (\underline{\mathcal{D}}(V) \otimes \underline{\mathcal{D}}(V)) &\to \underline{\mathcal{D}}(V, (U \otimes \underline{\mathcal{D}}(V)) \otimes V) \,, \\ u \otimes (f \otimes g) &\mapsto (v \mapsto (u \otimes f) \otimes g(v)) \end{aligned} \tag{3.23}$$

is such that the diagram

$$\begin{array}{ccc} U \otimes (\underline{\mathcal{D}}(V) \otimes \underline{\mathcal{D}}(V)) & \xrightarrow{U \otimes m} & U \otimes \underline{\mathcal{D}}(V) \\ \downarrow{\scriptstyle \phi_{U,V}} & & \downarrow{\scriptstyle \Phi_{U,V}} \\ \underline{\mathcal{D}}(V, (U \otimes \underline{\mathcal{D}}(V)) \otimes V) & \xrightarrow{\underline{\mathcal{D}}(V, U \otimes \mathrm{ev}_{V,V})} & \underline{\mathcal{D}}(V, U \otimes V) \end{array} \tag{3.24}$$

commutes. Indeed,

$$\begin{aligned} \Phi_{U,V} \circ (U \otimes m)(u \otimes a \otimes b) &= (v \mapsto u \otimes (a \circ b)(v)) \\ &= (v \mapsto u \otimes a(b(v))) \\ &= (v \mapsto u \otimes \mathrm{ev}_{V,V}(a \otimes b)(v)) \\ &= \underline{\mathcal{D}}(V, U \otimes \mathrm{ev}_{V,V}) \circ \phi_{U,V}(u \otimes a \otimes b) \end{aligned} \tag{3.25}$$

for all $u \in U$ and $a, b \in \underline{\mathcal{D}}(V)$. Naturality of the small braiding implies the right-hand side of (3.22) is equal to

$$\overrightarrow{\sigma}^{-1}_{\underline{\mathcal{D}}(V),U} \circ \Phi^{-1}_{U,V} \circ \underline{\mathcal{D}}(V, \overrightarrow{\sigma}_{V,U} \circ (\mathrm{ev}_{V,V} \otimes U) \circ \overleftarrow{\sigma}_{U,\underline{\mathcal{D}}(V)\otimes V}) \circ \phi_{U,V} \ . \tag{3.26}$$

Evaluating this on $u \otimes a \otimes b \in U \otimes (\underline{\mathcal{D}}(V) \otimes \underline{\mathcal{D}}(V))$ and using the hexagons (namely, (3.16)), we find

$$\begin{aligned}
u \otimes a \otimes b &\xmapsto{\phi_{U,V}} (v \mapsto (u \otimes a) \otimes b(v)) \\
&\xmapsto{\underline{\mathcal{D}}(V, \overleftarrow{\sigma}_{U,\underline{\mathcal{D}}(V)\otimes V})} (v \mapsto \mathcal{R}_2 a \otimes \mathcal{R}'_2 b(v) \otimes \mathcal{R}'_1 \mathcal{R}_1 u) \\
&\xmapsto{\underline{\mathcal{D}}(V, \mathrm{ev}_{V,V} \otimes U)} (v \mapsto \mathcal{R}_2 a(\mathcal{R}'_2 b(v)) \otimes \mathcal{R}'_1 \mathcal{R}_1 u) \\
&\xmapsto{\underline{\mathcal{D}}(V, \overrightarrow{\sigma}_{V,U})} (v \mapsto \mathcal{R}''_2 \mathcal{R}'_1 \mathcal{R}_1 u \otimes \mathcal{R}''_1 \mathcal{R}_2 a(\mathcal{R}'_2 b(v))) \\
&\xmapsto{\Phi^{-1}_{U,V}} \mathcal{R}''_2 \mathcal{R}'_1 \mathcal{R}_1 u \otimes \mathcal{R}''_1 \mathcal{R}_2 a(\mathcal{R}'_2 b(-)) \\
&\xmapsto{\overrightarrow{\sigma}^{-1}_{\underline{\mathcal{D}}(V),U}} \mathcal{R}_2 a(\mathcal{R}'_2 b(-)) \otimes \mathcal{R}'_1 \mathcal{R}_1 u \ .
\end{aligned} \tag{3.27}$$

Evaluating the left-hand side of (3.22) on $u \otimes a \otimes b$,

$$\begin{aligned}
u \otimes a \otimes b &\xmapsto{\overleftarrow{\sigma}_{U,\underline{\mathcal{D}}(V)\otimes\underline{\mathcal{D}}(V)}} \mathcal{R}_2 \cdot a \otimes \mathcal{R}_1 u \otimes b \\
&\xmapsto{\underline{\mathcal{D}}(V)\otimes\Phi_{U,V}} \mathcal{R}_2 \cdot a \otimes (v \mapsto \mathcal{R}_1 u \otimes b(v)) \\
&\xmapsto{\underline{\mathcal{D}}(V)\otimes\underline{\mathcal{D}}(V, \overrightarrow{\sigma}_{V,U} \overleftarrow{\sigma}_{U,V})} \mathcal{R}_2 \cdot a \otimes (v \mapsto \mathcal{R}''_2 \mathcal{R}'_1 \mathcal{R}_1 u \otimes \mathcal{R}''_1 \mathcal{R}'_2 b(v)) \\
&\xmapsto{\underline{\mathcal{D}}(V)\otimes\Phi^{-1}_{U,V}} \mathcal{R}_2 \cdot a \otimes \mathcal{R}''_2 \mathcal{R}'_1 \mathcal{R}_1 u \otimes \mathcal{R}''_1 \mathcal{R}'_2 b \\
&\xmapsto{\underline{\mathcal{D}}(V)\otimes\overrightarrow{\sigma}^{-1}_{\underline{\mathcal{D}},U})} \mathcal{R}_2 \cdot a \otimes \mathcal{R}'_2 b \otimes \mathcal{R}'_1 \mathcal{R}_1 u \\
&\xmapsto{m\otimes U} \mathcal{R}_2 a(\mathcal{R}'_2 b(-)) \otimes \mathcal{R}'_1 \mathcal{R}_1 u \ ,
\end{aligned} \tag{3.28}$$

which is equal to (3.27), as desired.

The proof that $\underline{\mathcal{D}}(M, -)$ is a braided module functor follows from an almost identical computation. All one needs do is to replace the left $\underline{\mathcal{D}}(M)$ strand in (3.22) with a $\underline{\mathcal{D}}(M)$-module, and apply the unit to the right strand. □

Remark 3.11. Let H be a quasitriangular Hopf algebra over a commutative ring K. Proposition 3.10 also holds for $\mathcal{D}$ being the category of H-modules and $\mathcal{C}$ full subcategory of rigid objects in $\mathcal{D}$, with the monoidal structure given by the tensor product over K. The proof is exactly the same.

It is well-known that $\mathcal{C}$ is a ribbon category, see for example [Ka95, Sec. XVII.3]. Furthermore, the coend $\mathcal{L} = \int^{V\in\mathcal{C}} V^* \otimes V$ exists in $\mathcal{D}$, and is given by the so-called *braided dual* [Kel23, pp. 26] (see also [BZBJ18a, Sec. 6] for the braided dual in a related setting). We then arrive at the following source of quantum moment maps coming from $U_\hbar\mathfrak{g}$-modules.

Theorem 3.12. Let $\mathcal{D}$ be the category of derived $I_\hbar$-adic complete $U_\hbar\mathfrak{g}$-modules, $\mathcal{C}$ the category of finite-dimensional $U_\hbar\mathfrak{g}$-modules, and $\underline{\mathcal{D}}(V) \in \mathcal{D}$ the internal endomorphism algebra of $V \in \mathcal{D}$. For all $V \in \mathcal{D}$, the morphism $\mu : \mathcal{L} \to \underline{\mathcal{D}}(V)$ defined via

$$\begin{array}{ccccc}
U^* \otimes U & \xrightarrow{(U^*\otimes U)\otimes u_V} & (U^* \otimes U) \otimes \underline{\mathcal{D}}(V) & \xrightarrow{\sim} & U^* \otimes \underline{\mathcal{D}}(V, U \otimes V) \\
\downarrow{\scriptstyle i_U} & & & & \\
\mathcal{L} & & & & \Big\downarrow{\scriptstyle U^*\otimes \underline{\mathcal{D}}(V, \overrightarrow{\sigma}_{V,U} \overleftarrow{\sigma}_{U,V})} \\
\vdots{\scriptstyle \mu} & & & & \\
\underline{\mathcal{D}}(V) & \xleftarrow[\mathrm{ev}_U \otimes \underline{\mathcal{D}}(V)]{} & (U^* \otimes U) \otimes \underline{\mathcal{D}}(V) & \xleftarrow[\sim]{} & U^* \otimes \underline{\mathcal{D}}(V, U \otimes V)
\end{array} \tag{3.29}$$

is a quantum moment map. Taking V to be a generalised Verma module, the quantum moment map $\mu : \mathcal{L} \to \underline{\mathcal{D}}(V)$ recovers the quantised conjugacy classes of Lie groups defined in [Mu07].

Proof. Applying Proposition 2.24 we immediately have that μ is a quantum moment map.

To relate this to the results of [Mu07], we begin by writing the explicit form of $\mu \circ i_U(f \otimes u)$ as an endomorphism of V. This is

$$\mu \circ i_U(f \otimes u) = (v \mapsto f(\mathcal{R}'_2\mathcal{R}_1 u)\mathcal{R}'_1\mathcal{R}_2 v) \, . \tag{3.30}$$

for all $v \in V$. Taking V to be a generalised Verma module, the map

$$\varphi : \mathcal{L} \to U_\hbar\mathfrak{g} \, , \quad i_U(f \otimes u) \mapsto f(\mathcal{R}'_2\mathcal{R}_1 u)\mathcal{R}'_1\mathcal{R}_2 \tag{3.31}$$

is precisely [Mu07, Eq. (20)]. Let $\rho_V : U_\hbar\mathfrak{g} \to \mathrm{End}_{\mathbb{C}[[\hbar]]}(V)$ be the $U_\hbar\mathfrak{g}$-module structure of V. The composition $\rho_V \circ \varphi$ is by definition [Mu07, Eq. (22)], and coincides with μ defined via (3.30). □

The representation theory of topologically free $U_\hbar\mathfrak{g}$ weight modules is analogous to the theory type-I $U_q^H\mathfrak{g}$-modules at generic q. That is to say, the examples in Section 3.2 make sense here as well (in particular, the tensor product decomposition of Proposition 3.7 follows from [Mu07, Prop. 3.7]). Moreover, we can now define quantum moment maps for these examples. Because of the decomposition in

(3.10), given any $\lambda \in \Lambda$ there exists $W \in \mathcal{C}$ and a surjection $W \otimes V(\lambda) \to V(\mu)$ for all $\mu \in \Lambda$, which implies $\mathcal{M}_\Lambda$ satisfies the conditions of $\mathcal{M}$ in Proposition A.7. Upon applying Proposition A.7 and Theorem 3.12 we obtain inclusions of braided module categories from $\mathcal{M}_\Lambda$:

Proposition 3.13. For all $\lambda \in \Lambda$, consider $\mathcal{D}_{A_\lambda}$ where $A_\lambda = \underline{\mathcal{D}}(V(\lambda))$ as a braided $\mathcal{C}$-module category, where the braiding is constructed from the quantum moment map obtained via Proposition 2.16. There is a full inclusion of braided module categories $\underline{\mathcal{D}}(V(\lambda), -) : \mathcal{M}_\Lambda \hookrightarrow \mathcal{D}_{A_\lambda}$.

Remark 3.14. The functor $\underline{\mathcal{D}}(V(\lambda), -)$ is not essentially surjective: For example, the full dual $\underline{\mathcal{D}}(V(\lambda), \mathbb{C}[[\hbar]])$ of the Verma module $V(\lambda)$ is not isomorphic to a direct sum of Verma modules. Hence $\underline{\mathcal{D}}(V(\lambda), -)$ is not an equivalence of braided module categories.

A Closed monoidal categories

We recall the definition and some basic facts about closed monoidal categories. We refer the reader to [EK65] for proofs of the general statements.

Definition A.1. [EK65, Sec. II.2 & Thm. II.5.3] A monoidal category $\mathcal{D}$ is *closed* if the functor $- \otimes V : \mathcal{D} \to \mathcal{D}$ has a right adjoint $\underline{\mathcal{D}}(V, -)$ for all $V \in \mathcal{D}$. That is, for all $U, V, W \in \mathcal{D}$ there is a natural isomorphism $\mathcal{D}(U \otimes V, W) \cong \mathcal{D}(U, \underline{\mathcal{D}}(V, W))$.

Let $\mathcal{D}$ be a closed monoidal category in the remainder of the appendix. Given $f \in \mathcal{D}(U \otimes V, W)$ we denote with $f' \in \mathcal{D}(U, \underline{\mathcal{D}}(V, W))$ the image of f under the isomorphism $\mathcal{D}(U \otimes V, W) \to \mathcal{D}(U, \underline{\mathcal{D}}(V, W))$. The internal-hom objects $\underline{\mathcal{D}}(V, W)$ come equipped with extra structure, which we now describe. The proofs of Lemmas A.2 and A.3 can be found in [EK65, Sec. II.3].

For each $V \in \mathcal{D}$, define the canonical natural transformations

$$\mathrm{ev}_{V,-} : \underline{\mathcal{D}}(V, -) \otimes V \Rightarrow - \quad \text{and} \quad \mathrm{coev}_{-,V} : - \Rightarrow \underline{\mathcal{D}}(V, - \otimes V) \tag{A.1}$$

to be the images of the identity morphisms under the isomorphisms

$$\mathcal{D}(\underline{\mathcal{D}}(V, -), \underline{\mathcal{D}}(V, -)) \Rightarrow \mathcal{D}(\underline{\mathcal{D}}(V, -) \otimes V, -) \,, \tag{A.2}$$

and

$$\mathcal{D}(- \otimes V, - \otimes V) \Rightarrow \mathcal{D}(-, \underline{\mathcal{D}}(V, - \otimes V)) \,, \tag{A.3}$$

respectively.

Lemma A.2. The natural transformations $\mathrm{ev}_{V,-}$ and $\mathrm{coev}_{-,V}$ are such that the diagrams

$$\begin{array}{ccc} U & \xrightarrow{f'} & \underline{\mathcal{D}}(V,W) \\ & \searrow^{\mathrm{coev}_{U,V}} & \uparrow{\scriptstyle \underline{\mathcal{D}}(V,f)} \\ & & \underline{\mathcal{D}}(V,U\otimes V) \end{array} \quad , \qquad \begin{array}{ccc} U\otimes V & \xrightarrow{f} & W \\ & \searrow^{f'\otimes V} & \uparrow{\scriptstyle \mathrm{ev}_{V,W}} \\ & & \underline{\mathcal{D}}(V,W)\otimes V \end{array} \tag{A.4}$$

commute for all $U,V,W \in \mathcal{D}$ and $f \in \mathcal{D}(U\otimes V, W)$.

For all $U,V,W,X \in \mathcal{D}$ there are canonical morphisms

$$m^W_{U,V} : \underline{\mathcal{D}}(U,V)\otimes\underline{\mathcal{D}}(W,U) \to \underline{\mathcal{D}}(W,V)\ , \tag{A.5}$$

given by the image of

$$\mathrm{ev}_{U,V}\circ(\underline{\mathcal{D}}(U,V)\otimes \mathrm{ev}_{W,U}) : \underline{\mathcal{D}}(U,V)\otimes(\underline{\mathcal{D}}(W,U)\otimes W) \to V \tag{A.6}$$

under the isomorphism

$$\mathcal{D}(\underline{\mathcal{D}}(U,V)\otimes(\underline{\mathcal{D}}(W,U)\otimes W),V) \to \mathcal{D}(\underline{\mathcal{D}}(U,V)\otimes\underline{\mathcal{D}}(W,U),\underline{\mathcal{D}}(W,V))\ . \tag{A.7}$$

Lemma A.3. The morphisms $m^W_{U,V}$ are such that the diagram

$$\begin{array}{ccccc} & & \underline{\mathcal{D}}(U,V)\otimes\underline{\mathcal{D}}(X,U) & & \\ & {\scriptstyle \underline{\mathcal{D}}(U,V)\otimes m^X_{W,U}}\nearrow & & \searrow{\scriptstyle m^X_{U,V}} & \\ \underline{\mathcal{D}}(U,V)\otimes(\underline{\mathcal{D}}(W,U)\otimes\underline{\mathcal{D}}(X,W)) & & & & \underline{\mathcal{D}}(X,V) \\ {\scriptstyle\sim}\downarrow & & & & \uparrow{\scriptstyle m^X_{W,V}} \\ (\underline{\mathcal{D}}(U,V)\otimes\underline{\mathcal{D}}(W,U))\otimes\underline{\mathcal{D}}(X,W) & & \xrightarrow[m^W_{U,V}\otimes\underline{\mathcal{D}}(X,W)]{} & & \underline{\mathcal{D}}(W,V)\otimes\underline{\mathcal{D}}(X,W) \end{array} \tag{A.8}$$

commutes.

Denote with u_V the composition

$$\mathbb{1} \xrightarrow{\mathrm{coev}_{\mathbb{1},V}} \underline{\mathcal{D}}(V,\mathbb{1}\otimes V) \xrightarrow{\underline{\mathcal{D}}(V,l_V)} \underline{\mathcal{D}}(V,V) =: \underline{\mathcal{D}}(V)\ . \tag{A.9}$$

Using Lemmas A.2 and A.3 we construct algebras and bimodules in $\mathcal{D}$ from the internal-hom objects:

Corollary A.4. For all $U,V \in \mathcal{D}$, each tuple $(\underline{\mathcal{D}}(V), m^V_{V,V}, u_V)$ is an algebra in $\mathcal{D}$, which we refer to simply as $\underline{\mathcal{D}}(V)$. Each tuple $(\underline{\mathcal{D}}(U,V), m^V_{U,V}, m^V_{U,V})$ is a $(\underline{\mathcal{D}}(V),\underline{\mathcal{D}}(U))$-bimodule in $\mathcal{D}$.

Define the natural transformation $\Phi^{\equiv}_{-,=} : - \otimes \underline{\mathcal{D}}(\equiv, =) \to \underline{\mathcal{D}}(\equiv, - \otimes =)$ via the composition

$$\Phi^{U}_{V,W} : V \otimes \underline{\mathcal{D}}(U,W) \xrightarrow{\mathrm{coev}_{V,W} \otimes \underline{\mathcal{D}}(U,W)} \underline{\mathcal{D}}(W, V \otimes W) \otimes \underline{\mathcal{D}}(U,W) \xrightarrow{m^{U}_{W,V\otimes W}} \underline{\mathcal{D}}(U, V \otimes W) \ . \tag{A.10}$$

The following lemma is an approximation to [EGNO15, Lem. 7.9.4] in our setting.

Lemma A.5. The natural transformation $\Phi^{=}_{V,-}$ is an isomorphism if and only if V has a left-dual in $\mathcal{D}$.

Proof. Let $U, V, W \in \mathcal{D}$ and V be rigid. For all $X \in \mathcal{D}$, we have

$$\begin{aligned} \mathcal{D}(X \otimes U, V \otimes W) &\cong \mathcal{D}(V^* \otimes (X \otimes U), W) \\ &\cong \mathcal{D}((V^* \otimes X) \otimes U, W) \\ &\cong \mathcal{D}(V^* \otimes X, \underline{\mathcal{D}}(U,W)) \\ &\cong \mathcal{D}(X, V \otimes \underline{\mathcal{D}}(U,W)) \ . \end{aligned} \tag{A.11}$$

By definition of the internal-hom, we also have

$$\mathcal{D}(X \otimes U, V \otimes W) \cong \mathcal{D}(X, \underline{\mathcal{D}}(U, V \otimes W)) \ . \tag{A.12}$$

Hence, by the Yoneda Lemma, we have $V \otimes \underline{\mathcal{D}}(U,W) \cong \underline{\mathcal{D}}(U, V \otimes W)$. Furthermore, this isomorphism exhibits $V \otimes \underline{\mathcal{D}}(U,W)$ as representing $\mathcal{D}(- \otimes U, V \otimes W)$, making it exactly $\Phi^{U}_{V,W}$.

Assume $\Phi^{=}_{V,-}$ is an isomorphism. Let $U, V, W \in \mathcal{D}$ and consider the object $\underline{\mathcal{D}}(V, \mathbb{1})$. We have evaluation and coevaluation morphisms given by

$$\mathrm{ev}_V := \mathrm{ev}_{V,\mathbb{1}} : \underline{\mathcal{D}}(V, \mathbb{1}) \otimes V \to \mathbb{1} \tag{A.13}$$

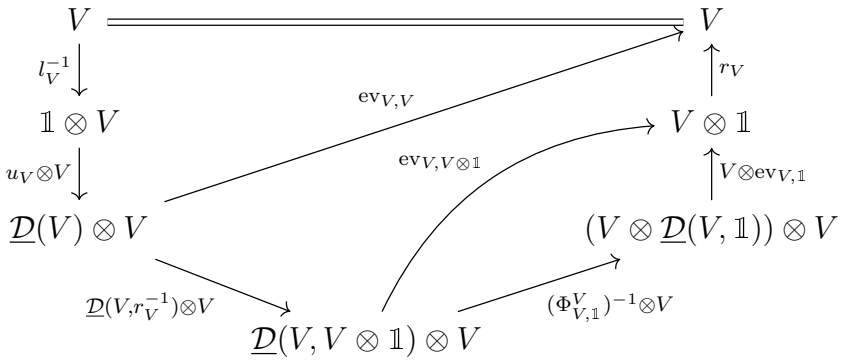
and

$$\mathrm{coev}_V : \mathbb{1} \xrightarrow{u_V} \underline{\mathcal{D}}(V) \xrightarrow{\underline{\mathcal{D}}(V, r_V^{-1})} \underline{\mathcal{D}}(V, V \otimes \mathbb{1}) \xrightarrow{(\Phi^{V}_{V,\mathbb{1}})^{-1}} V \otimes \underline{\mathcal{D}}(V, \mathbb{1}) \ . \tag{A.14}$$

We need to check the zig-zag identities. For brevity we will suppress the associators in what follows. We will check the identity

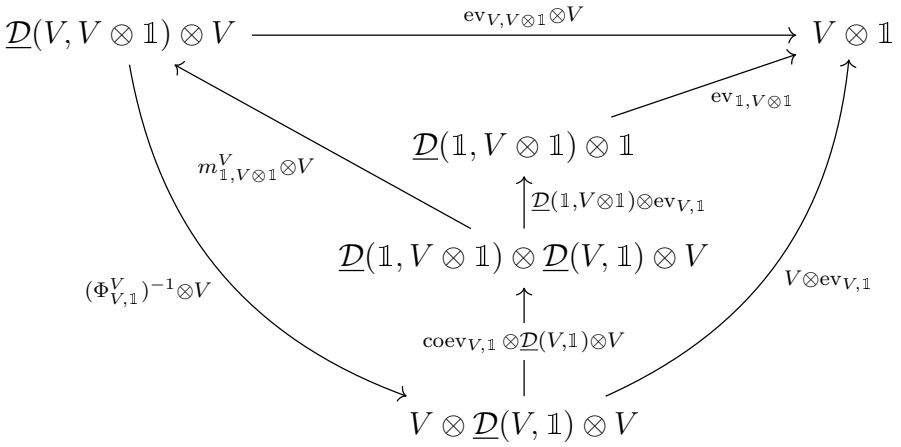
$$\begin{array}{ccc} V & = & V \\ {\scriptstyle l_V^{-1}}\downarrow & & \uparrow{\scriptstyle r_V} \\ \mathbb{1} \otimes V & & V \otimes \mathbb{1} \\ & {\scriptstyle \mathrm{coev}_V \otimes V} \searrow \quad \nearrow {\scriptstyle V \otimes \mathrm{ev}_V} & \\ & (V \otimes \underline{\mathcal{D}}(V, \mathbb{1})) \otimes V & \end{array} \ . \tag{A.15}$$

The other identity follows via a similar computation. The diagram (A.15) is equivalent to the outside pentagon of



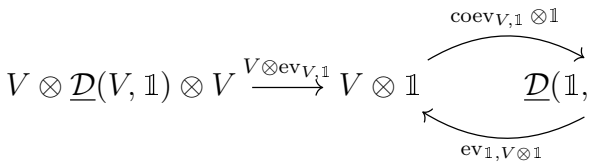
(A.16)

The square in the top left of (A.16) commutes from the second diagram in Lemma A.2 and the definition of u_V, and the square in the middle from the naturality of $\mathrm{ev}_{V,-}$. It remains to show that the bottom right triangle in (A.16) commutes. This triangle is the outside of the diagram



(A.17)

The triangle on the bottom left of (A.17) commutes by the definition of $\Phi^V_{V,\mathbb{1}}$, and the upper left square of (A.17) commutes from the definition of $m^V_{\mathbb{1},V\otimes\mathbb{1}}$ and the second diagram in Lemma A.2. The remaining square in (A.17) can be reshaped into



(A.18)

The loop in (A.18) is a special case of [EK65, Eq. (3.6)], so (A.17) commutes. □

Assume $\mathcal{D}$ is abelian and has a right-exact tensor product. Let $\mathcal{C} \subset \mathcal{D}$ be a full rigid monoidal subcategory of $\mathcal{D}$. For the remainder of the subsection, fix a full subcategory $\mathcal{M} \subset \mathcal{D}$ which is closed under tensoring with objects in $\mathcal{C}$, making $\mathcal{M}$ a $\mathcal{C}$-module category. Given $M \in \mathcal{M}$, we consider the algebra $\underline{\mathcal{D}}(M)$ and the $\mathcal{D}$-module category $\mathcal{D}_{\underline{\mathcal{D}}(M)}$. By virtue of the fact that $\mathcal{C}$ is a monoidal subcategory of $\mathcal{D}$, the category $\mathcal{D}_{\underline{\mathcal{D}}(M)}$ is also a $\mathcal{C}$-module category. Consider the functor $\mathcal{D}(M, -) : \mathcal{M} \to \mathcal{D}_{\underline{\mathcal{D}}(M)}$. The following is a corollary of Lemma A.5

Lemma A.6. Assume $\mathcal{D}$ is abelian and has a right-exact tensor product. Given $M \in \mathcal{M}$, the natural isomorphism $\Phi^M_{-,=}$ gives the functor

$$\underline{\mathcal{D}}(M, -) : \mathcal{M} \to \mathcal{D}_{\underline{\mathcal{D}}(M)} , \quad N \mapsto \underline{\mathcal{D}}(M, N) \tag{A.19}$$

the structure of a $\mathcal{C}$-module functor.

The following proposition is in the spirit of the characterisation given in [EGNO15, Thm. 7.10.1] in the non-finite and non-rigid setting. The proof is exactly that in steps (1) and (2) of the proof given for [EGNO15, Thm. 7.10.1], however our assumptions differ so we repeat it here.

Proposition A.7. Let there be an $M \in \mathcal{M}$, such that for each $N \in \mathcal{M}$ there exists $U \in \mathcal{C}$ and an epimorphism $U \otimes M \to N$. Assume also that $\underline{\mathcal{D}}(M, -)$ is right exact. Then the $\mathcal{C}$-module functor $\underline{\mathcal{D}}(M, -) : \mathcal{M} \to \mathcal{D}_{\underline{\mathcal{D}}(M)}$ is fully faithful.

Proof. Assume $\mathcal{D}$ is abelian and has a right-exact tensor product. Say $N \in \mathcal{M}$ is of the form $U \otimes M$ for some $U \in \mathcal{C}$. Given $L \in \mathcal{M}$, we have

$$\begin{aligned} \mathcal{D}_{\underline{\mathcal{D}}(M)}(\underline{\mathcal{D}}(M, U \otimes M), \underline{\mathcal{D}}(M, L)) &\cong \mathcal{D}_{\underline{\mathcal{D}}(M)}(U \otimes \underline{\mathcal{D}}(M), \underline{\mathcal{D}}(M, L)) \\ &\cong \mathcal{D}(U, \underline{\mathcal{D}}(M, L)) \\ &\cong \mathcal{M}(U \otimes M, L) \\ &= \mathcal{M}(N, L) , \end{aligned} \tag{A.20}$$

where the second line follows from the fact that $\mathcal{D}_A(X \otimes A, B) \cong \mathcal{D}(A, B)$ for all $X \in \mathcal{D}$, and any algebra $A \in \mathcal{D}$, and all $B \in \mathcal{D}_A$.

The condition on $\mathcal{M}$ and the fact that $F := \underline{\mathcal{D}}(M, -)$ is right exact implies that for all $N \in \mathcal{M}$ there are $U, W \in \mathcal{D}$ such that each row of the diagram

$$\begin{array}{ccccccc} U \otimes M & \longrightarrow & W \otimes M & \longrightarrow & N & \longrightarrow & 0 \\ {\scriptstyle F}\downarrow & & {\scriptstyle F}\downarrow & & {\scriptstyle F}\downarrow & & \\ F(U \otimes M) & \longrightarrow & F(W \otimes M) & \longrightarrow & F(N) & \longrightarrow & 0 \end{array} \tag{A.21}$$

is exact. Because $\mathcal{M}$ is a full subcategory of $\mathcal{D}$, and $\mathcal{D}$ is abelian, the (contravariant) hom functors $\mathcal{M}(-, L)$ are left exact for all $L \in \mathcal{M}$. Define H to be the functor $\mathcal{D}_{\underline{\mathcal{D}}(M)}(F(-), F(L))$. Hence each row of the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathcal{M}(N, L) & \longrightarrow & \mathcal{M}(W \otimes M, L) & \longrightarrow & \mathcal{M}(U \otimes M, L) \\ & & \downarrow{\scriptstyle F} & & \downarrow{\scriptstyle F} & & \downarrow{\scriptstyle F} \\ 0 & \longrightarrow & H(N) & \longrightarrow & H(W \otimes M) & \longrightarrow & H(U \otimes M) \end{array} \tag{A.22}$$

is exact. From (A.20) we know that the second and third vertical arrows are isomorphisms, hence the first vertical arrow is an isomorphism. Thus $F = \underline{\mathcal{D}}(M, -)$ is fully-faithful. □

Remark A.8. An obstruction to $\underline{\mathcal{D}}(V, -)$ being an equivalence of $\mathcal{C}$-module categories is that if $\mathcal{M}$ is not equivalent to $\mathcal{D}$ the functor $\underline{\mathcal{D}}(M, -)$ cannot be essentially surjective. To see this, consider $\underline{\mathcal{D}}(M, W) \in \mathcal{D}_{\underline{\mathcal{D}}(M)}$ where $W \notin \mathcal{M}$.

B Derived I-adic completion

Let R be a commutative ring and I an ideal of R. In this Appendix we review the notion of derived I-adic completions of R-modules from [PW22]. The key point is that the category of derived I-adic complete modules is an abelian monoidal category with right-exact tensor product. We omit proofs, and refer the reader to [PW22] and [Ard04] a detailed discussion.

First we review the notion of I-adic completion which usually arises in the study of quantum groups.

Definition B.1. [Ka95, Sec. XVI.9] Let $(M_n)_{n\geq 0}$ be a family of R-modules and $(\rho_n : M_n \to M_{n-1})_{n>0}$ a family of R-linear maps. The *inverse limit of* $(M_n)_n$ is the R-module

$$\varprojlim_n M_n := \{(m_n)_{n\geq 0} \in \prod_{n\geq 0} M_n : \rho_n(m_n) = m_{n-1} \text{ for all } n > 0\} , \tag{B.1}$$

with the module and group structure defined component-wise: $r \cdot (a_n) = (ra_n)$, $(a_n)_n + (b_n)_n = (a_n + b_n)_n$.

Definition B.2. [PW22, Def. 2.1] Let M be an R-module, $(M_n)_{n\geq 0}$ be the family of submodules $M_n = M/I^n M$, and π_n the canonical projections

$$\pi_n : M_n \to M_{n-1} . \tag{B.2}$$

The *I-adic completion of M* is the inverse limit $\widehat{M} = \varprojlim_n M/I^nM$. The module $\widehat{M}$ comes with a canonical R-linear map $i : M \to \widehat{M}$, and if i is an isomorphism M is said to be *I-adic complete*. Given two R-modules M and N their *completed tensor product* $M\widehat{\otimes}N$ is defined to be the I-adic completion of $M \otimes_R N$. The category of I-adic complete R-modules is the full subcategory of R-modules in which every object is I-adic complete.

Functoriality of I-adic completion follows from the arguments in [Ka95, Sec. XVI.9]. Let $\widehat{(-)} : R - \mathsf{mod} \to R - \mathsf{mod}$ be the completion functor and $\widehat{(-)}_0$ its zeroth left-derived functor.

Definition B.3. [PW22, Def. 3.1] The *derived I-adic completion* $\widehat{M}_0$ of an R-module M is the image of M upon applying the functor $\widehat{(-)}_0$. M is said to be *derived I-adic complete* if the canonical morphism $M \to \widehat{M}_0$ is an isomorphism. Given two R-modules M and N their *derived completed tensor product* $M\widehat{\otimes}_0N$ is the derived I-adic completion of $M \otimes N$.

Proposition B.4. [PW22, Prop.s 3.9 & 3.10] The category of derived I-adic complete R-modules is a symmetric closed abelian monoidal category containing all limits and colimits.

Corollary B.5. [Ard04] Let $\mathcal{C}$ be the category of derived I-adic complete R-modules. Given an algebra $A \in \mathcal{C}$, the category of A-modules in $\mathcal{C}$ is abelian.

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